# Software Architecture

17-313: Foundations of Software Engineering

https://cmu-313.github.io

**Michael Hilton** and Josh Sunshine

Fall 2026

# Administrivia

- Midterm graded and grades posted

Midterm 1 80.0 points

| Minimum | Median | Maximum | Mean | Std Dev |
|---|---|---|---|---|
| 62.5% | 82.5% | 97.5% | 81.37% | 8.55% |

  - For comparison, last semester:

Midterm 1 80.0 points

| Minimum | Median | Maximum | Mean | Std Dev |
|---|---|---|---|---|
| 50.0% | 85.63% | 97.5% | 81.65% | 11.99% |

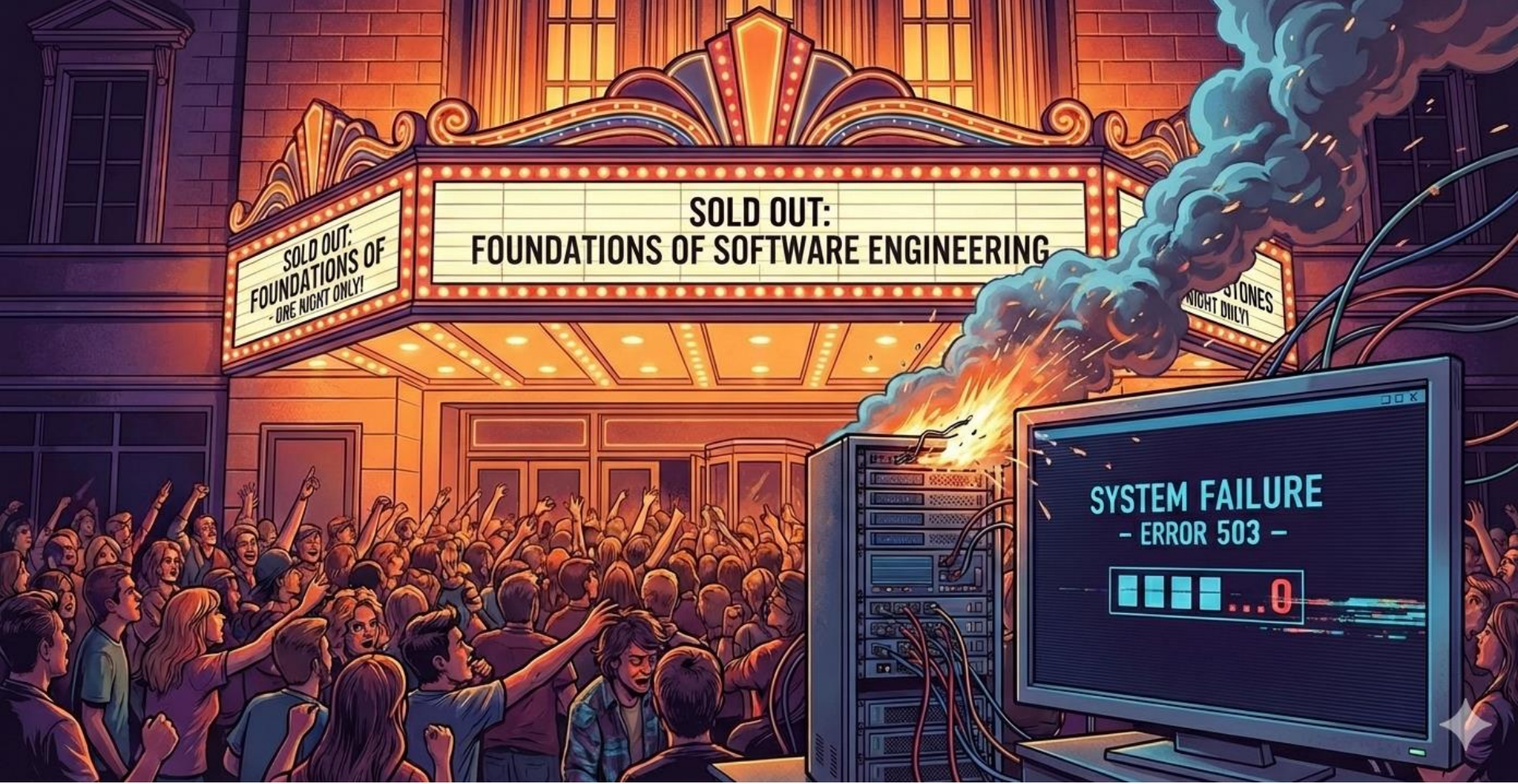
  - P3 has not be released yet, hope to release tomorrow. Problem set is available for you to work on.
  - Midsemester grades released tomorrow (Wednesday)

# Case Study: Victim of Success Theater

# Case Study: Victim of Success Theater

- Simple web application for a local theater:
  - Sell tickets
  - 500 seats
- Architecture: Standard "3-Tier" setup:
  UI → API → Database
- Happy path :
  - A user clicks "Buy”
  - API checks the DB, decrements the ticket count, and sends a confirmation
  - It works perfectly for months.




SOLD OUT:
FOUNDATIONS OF SOFTWARE ENGINEERING
SOLD OUT:
FOUNDATIONS OF
- ONE NIGHT ONLY!
SYSTEM FAILURE
- ERROR 503 -

# In-class activity

**Where does the simple architecture of ”Victim of Success Theater” break down?**

1. Think on your own
2. Write down a hypothesis or two on your participation sheet
3. Discuss with your neighbor
4. Share with the class

# Case Study: Quality Attributes

- **Scalability:** Can we handle the 10:00 AM rush?
- **Reliability:** Can we fail gracefully without double-charging people?
- **Maintainability:** Can we swap adjust the computing resources dedicated to the app without rewriting the code?

# Software Architecture

*The software architecture of a program or computing system is the structure or structures of the system, which comprise software elements, the externally visible properties of those elements, and the relationships among them.*

*[Bass et al. 2003]*

This definition is ambivalent to whether the architecture is known or whether it's any good!

# Software Architecture: Motivation

- Facilitates internal and external communication
- Describes design decisions and prescribes implementation constraints
- Relates to organizational structure
- Permits/precludes achieving non-functional requirements
- Control complexity
- Reason about and manage change
- Good basis for effort estimation
- …

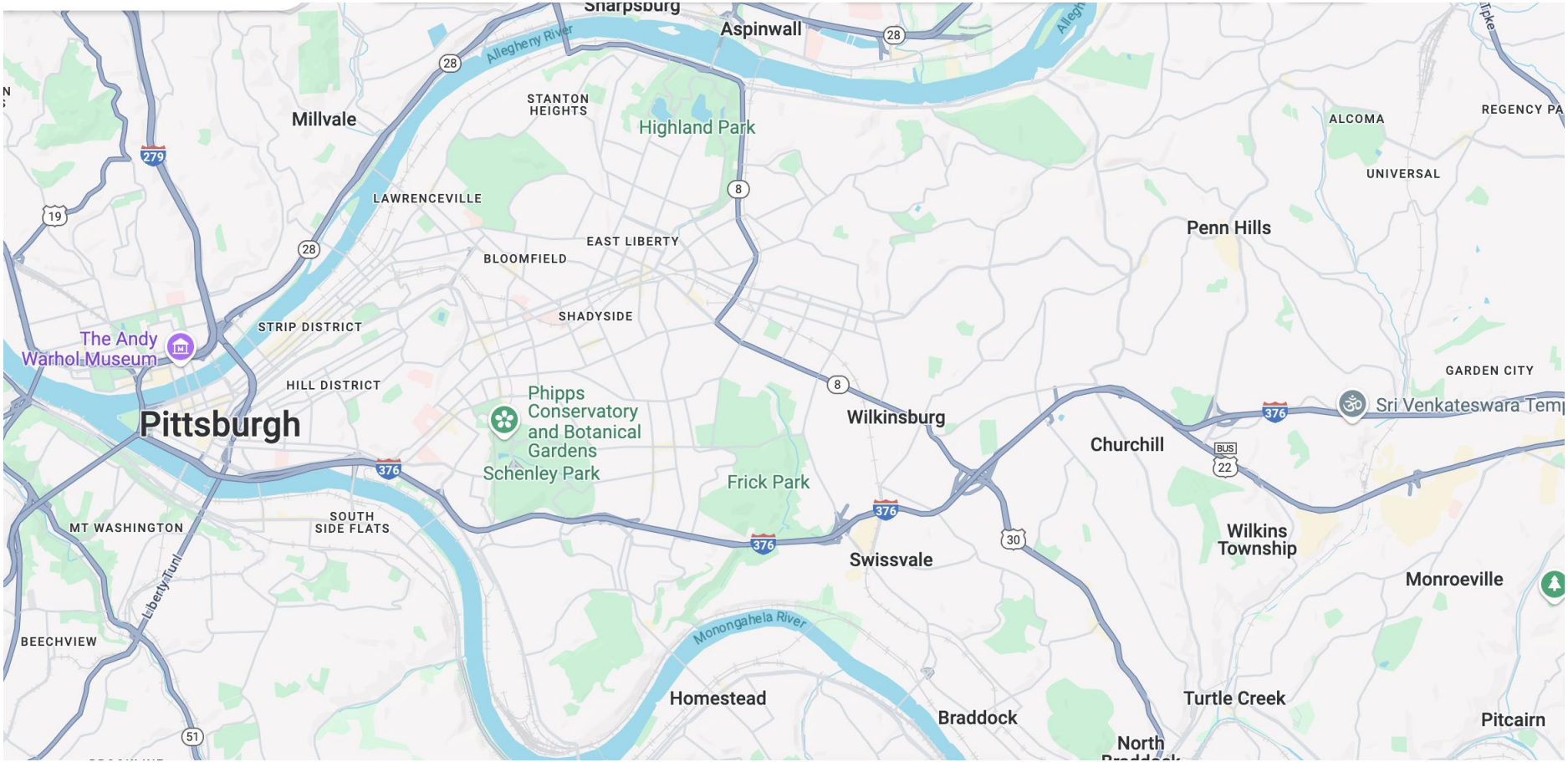
*Architecting Software the SEI Way - Software Architecture Fundamentals: Technical, Business, and Social Influences.* Robert Wojcik. 2012

# Conway's Law

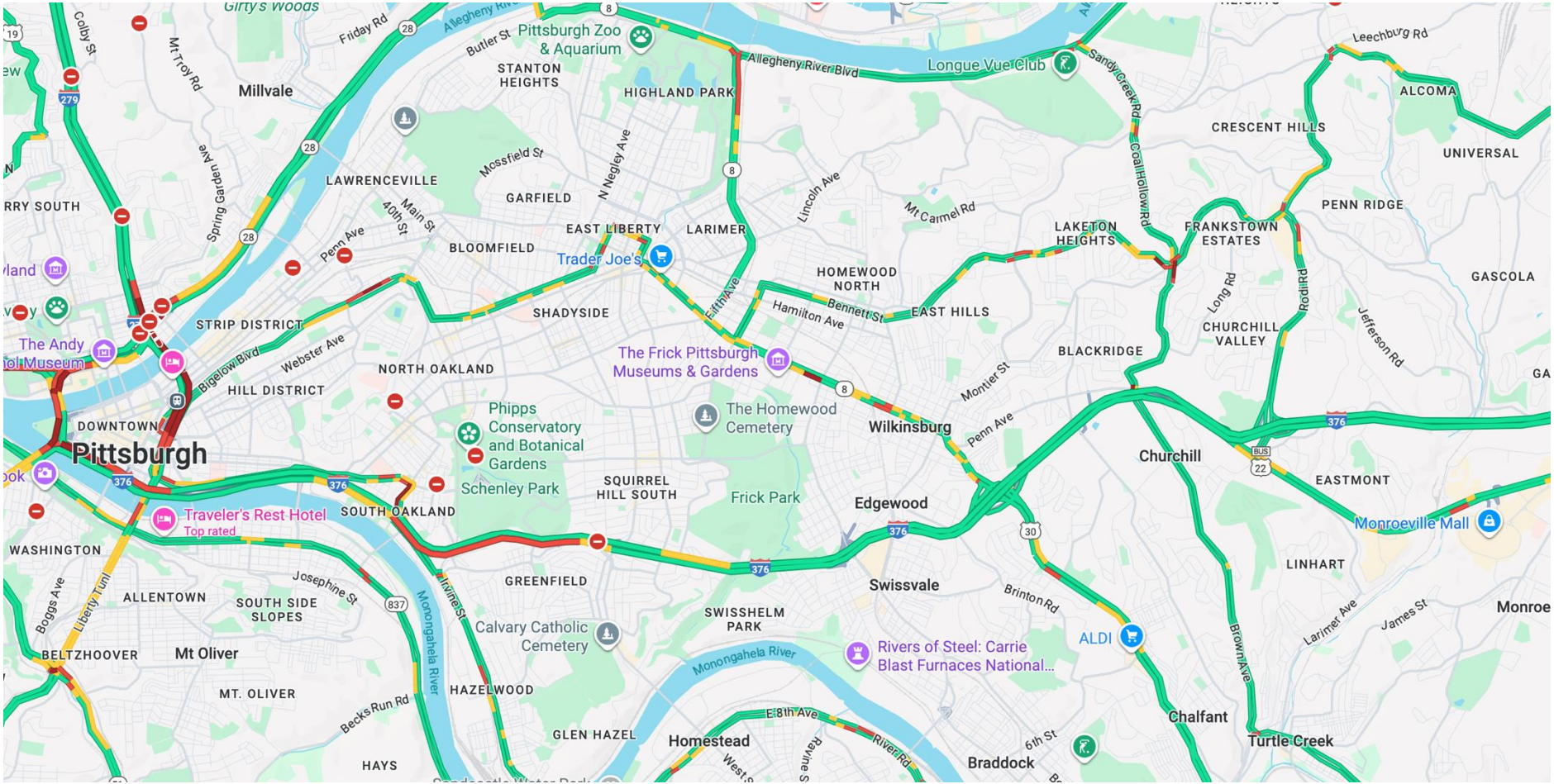


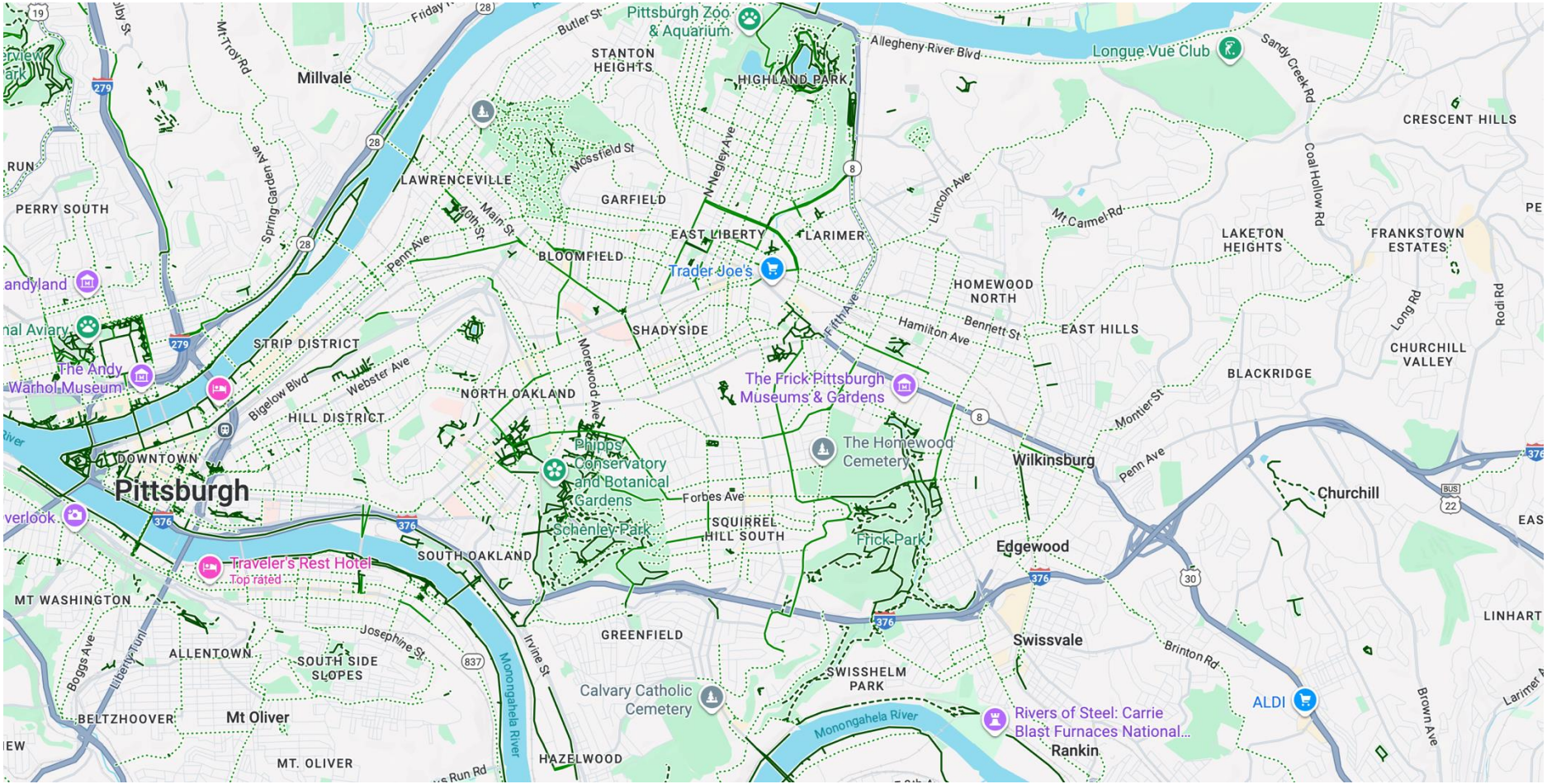

Company Structure **influences** Software Architecture

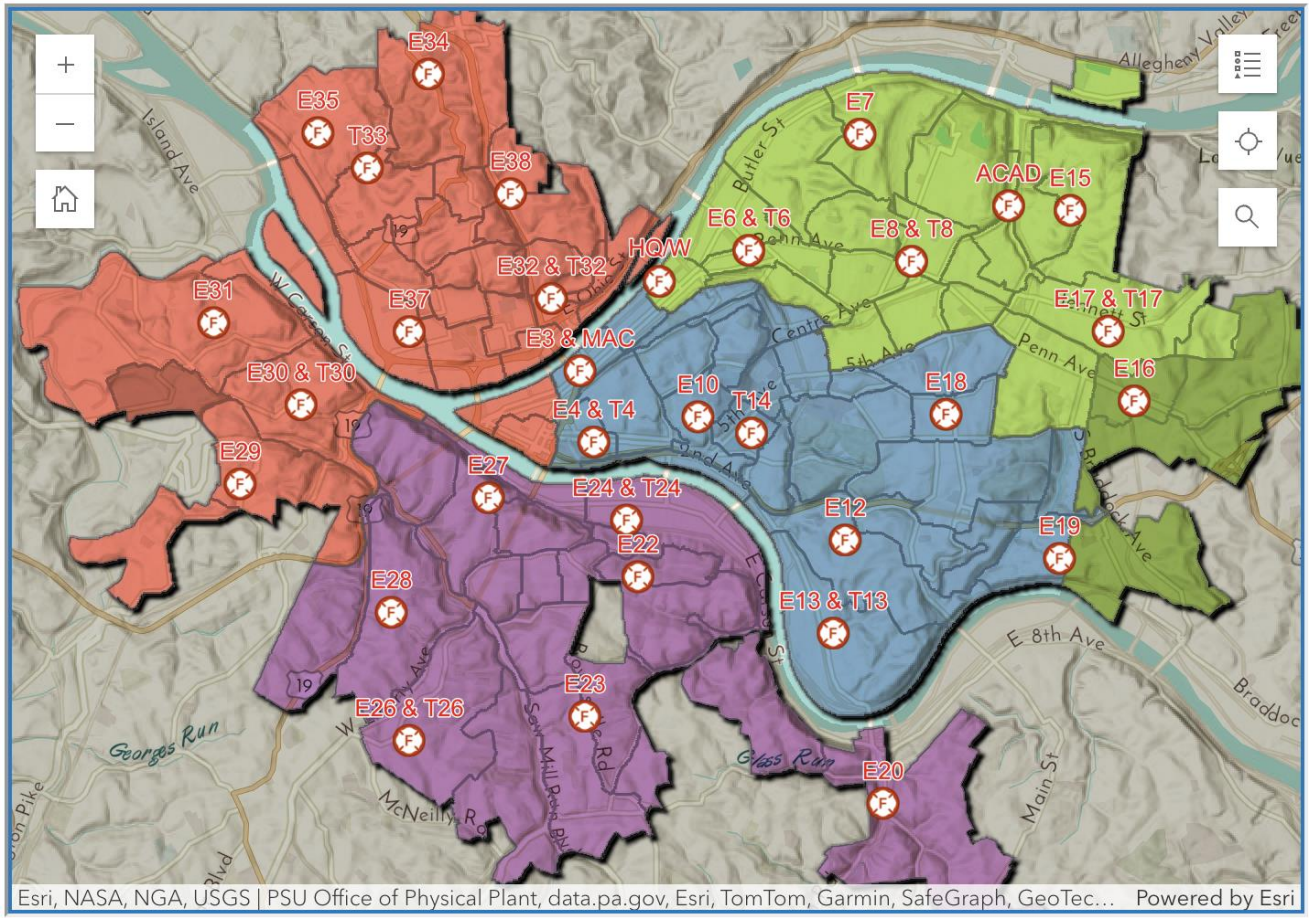
# Architectural Views


Sharpsburg
Aspinwall
Allegheny River
28
Millvale
STANTON HEIGHTS
Highland Park
ALCOMA
REGENCY PA
UNIVERSAL
279
19
LAWRENCEVILLE
8
EAST LIBERTY
Penn Hills
BLOOMFIELD
SHADYSIDE
STRIP DISTRICT
The Andy Warhol Museum
HILL DISTRICT
Pittsburgh
Phipps Conservatory and Botanical Gardens
Schenley Park
Wilkinsburg
GARDEN CITY
Sri Venkateswara Temp
376
Churchill
BUS 22
Frick Park
MT WASHINGTON
SOUTH SIDE FLATS
Liberty Tunl
Swissvale
30
Wilkins Township
Monroeville
BEECHVIEW
Monongahela River
Homestead
Braddock
Turtle Creek
Pitcairn
North
51


Aspinwall
Allegheny River
Millvale
STANTON HEIGHTS
HIGHLAND PARK
ALCOMA
UNIVERSAL
LAWRENCEVILLE
EAST LIBERTY
BLOOMFIELD
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North Braddock
GARDEN CIT
Sri Venkateswara Te
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51
BUS


Girty's Woods
Colby St
Mt Troy Rd
Millvale
Friday Rd
Allegheny River
Butler St
Pittsburgh Zoo & Aquarium
STANTON HEIGHTS
HIGHLAND PARK
Allegheny River Blvd
Longue Vue Club
Sandy Creek Rd
Leechburg Rd
ALCOMA
CRESCENT HILLS
UNIVERSAL
Spring Garden Ave
LAWRENCEVILLE
Mossfield St
N Negley Ave
GARFIELD
Main St
40th St
Penn Ave
BLOOMFIELD
EAST LIBERTY
LARIMER
Lincoln Ave
Mt Carmel Rd
LAKETON HEIGHTS
Coal Hollow Rd
FRANKSTOWN ESTATES
PENN RIDGE
GASCOLA
Trader Joe's
HOMEWOOD NORTH
Fifth Ave
Hamilton Ave
Bennett St
EAST HILLS
Long Rd
Rodi Rd
Jefferson Rd
CHURCHILL VALLEY
STRIP DISTRICT
SHADYSIDE
The Andy Warhol Museum
Bigelow Blvd
Webster Ave
NORTH OAKLAND
The Frick Pittsburgh Museums & Gardens
BLACKRIDGE
Montier St
HILL DISTRICT
DOWNTOWN
Pittsburgh
Phipps Conservatory and Botanical Gardens
The Homewood Cemetery
Wilkinsburg
Penn Ave
Churchill
EASTMONT
Schenley Park
SQUIRREL HILL SOUTH
Frick Park
Edgewood
Traveler's Rest Hotel
Top rated
SOUTH OAKLAND
Monroeville Mall
WASHINGTON
LINHART
GREENFIELD
Swissvale
Brinton Rd
Josephine St
Boggs Ave
Liberty Tunl
ALLENTOWN
SOUTH SIDE SLOPES
Monongahela River
Calvary Catholic Cemetery
SWISSHELM PARK
Larimer Ave
James St
Monroe
ALDI
BELTZHOOVER
Mt Oliver
Rivers of Steel: Carrie Blast Furnaces National...
Monongahela River
Brown Ave
MT. OLIVER
HAZELWOOD
Becks Run Rd
E 8th Ave
Chalfant
GLEN HAZEL
Homestead
6th St
River Rd
Turtle Creek
HAYS
Braddock


Pittsburgh Zoo & Aquarium
Allegheny River Blvd
Longue Vue Club
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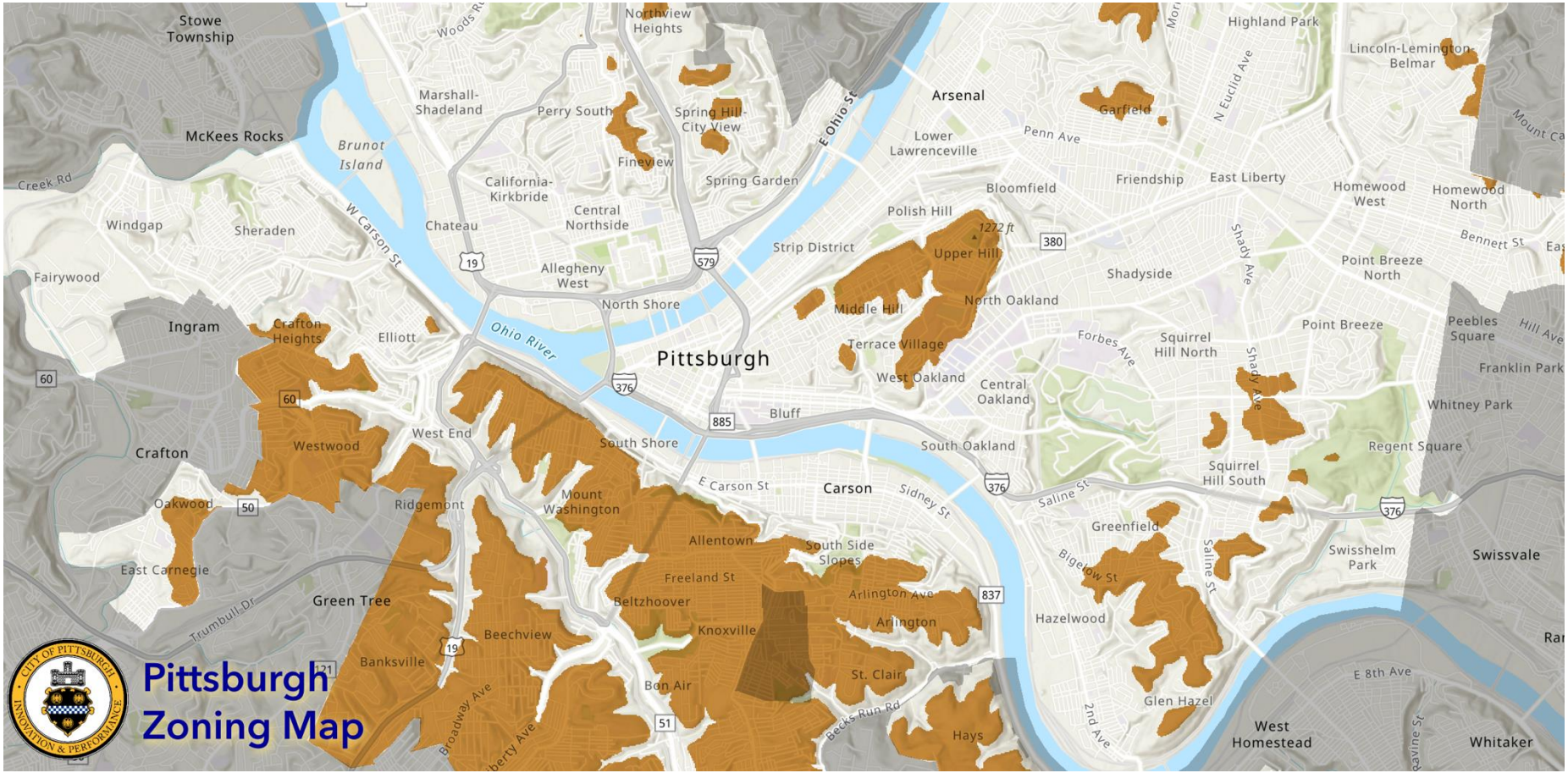



**Fire Station**

**Fire Districts**

1
2
3
4


Zoning
Zoning Districts
Uptown Public Realm
Riverfront
Specially Planned
Planned Unit Development
Educational/Medical Institution
Neighborhood Office
Parks
Hillside
General Industrial
Neighborhood Industrial
Urban Industrial
Highway Commercial
Local Neighborhood Commercial
Urban Neighborhood Commercial
Urban Center Employment
Urban Center Mixed Use
Single-Unit Attached Residential
Single-Unit Detached Residential
Two-Unit Residential
Three-Unit Residential
Pittsburgh
Central Northside
Allegheny Center
Allegheny West
Strip District
Polish Hill
Bloomfield
Upper Hill
Bedford Dwellings
Middle Hill
North Oakland
Shadyside
UPMC Shadyside
Acrisure Stadium
North Shore
Point State Park
Downtown Pittsburgh
Crawford-Roberts
Terrace Village
University of Pittsburgh
Carnegie Mellon University
Squirrel Hill North
West Oakland
Central Oakland
The Bob O'Connor Golf Course at Schenley Park
Schenley Park
Duquesne University
Bluff
South Shore
South Oakland
Carson
Mount
Allentown
South Side Slopes
Greenfield

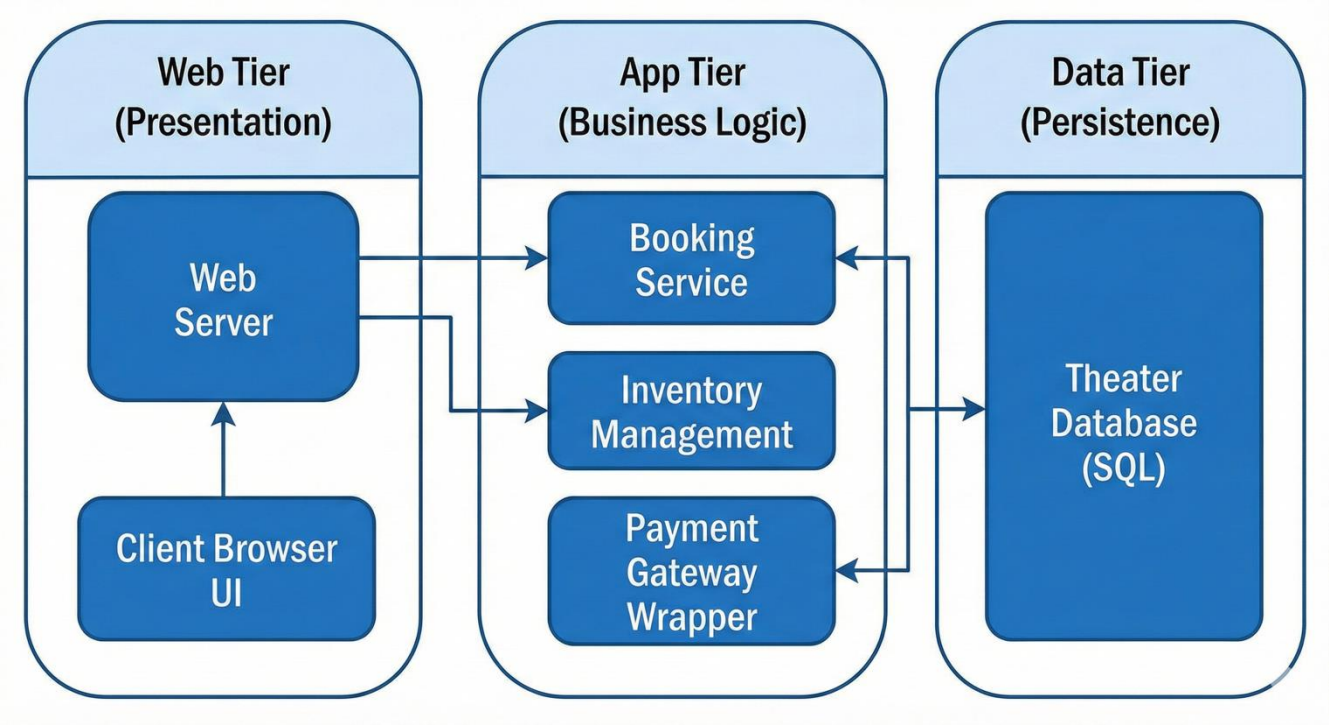

Pittsburgh
Zoning Map
Stowe Township
McKees Rocks
Brunot Island
Windgap
Sheraden
Fairywood
Ingram
Crafton Heights
Elliott
Crafton
Westwood
Oakwood
East Carnegie
Green Tree
Trumbull Dr
Banksville
Beechview
Broadway Ave
Liberty Ave
Marshall-Shadeland
Perry South
Northview Heights
Spring Hill-City View
Fineview
California-Kirkbride
Central Northside
Chateau
Spring Garden
Allegheny West
North Shore
Ohio River
W Carson St
E Ohio St
Pittsburgh
West End
South Shore
Mount Washington
Ridgemont
Allentown
Beltzhoover
Knoxville
Bon Air
Arsenal
Lower Lawrenceville
Garfield
Penn Ave
Strip District
Polish Hill
1272 ft
Upper Hill
Middle Hill
Terrace Village
West Oakland
Central Oakland
Bloomfield
Friendship
East Liberty
Homewood West
Homewood North
Shadyside
Point Breeze North
Point Breeze
North Oakland
Squirrel Hill North
Squirrel Hill South
Forbes Ave
Shady Ave
Bennett St
Peebles Square
Franklin Park
Whitney Park
Regent Square
Bluff
South Oakland
Carson
E Carson St
Sidney St
Saline St
South Side Slopes
Arlington Ave
Arlington
Freeland St
Hazelwood
Greenfield
Bigelow St
Swisshelm Park
Swissvale
Glen Hazel
E 8th Ave
2nd Ave
Becks Run Rd
Hays
West Homestead
Whitaker
Highland Park
Lincoln-Lemington-Belmar
N Euclid Ave
Mount Ca
Woods R
Creek Rd
60
19
579
376
885
50
121
51
380
837

# Each views abstracts reality to focus on conveying specific information

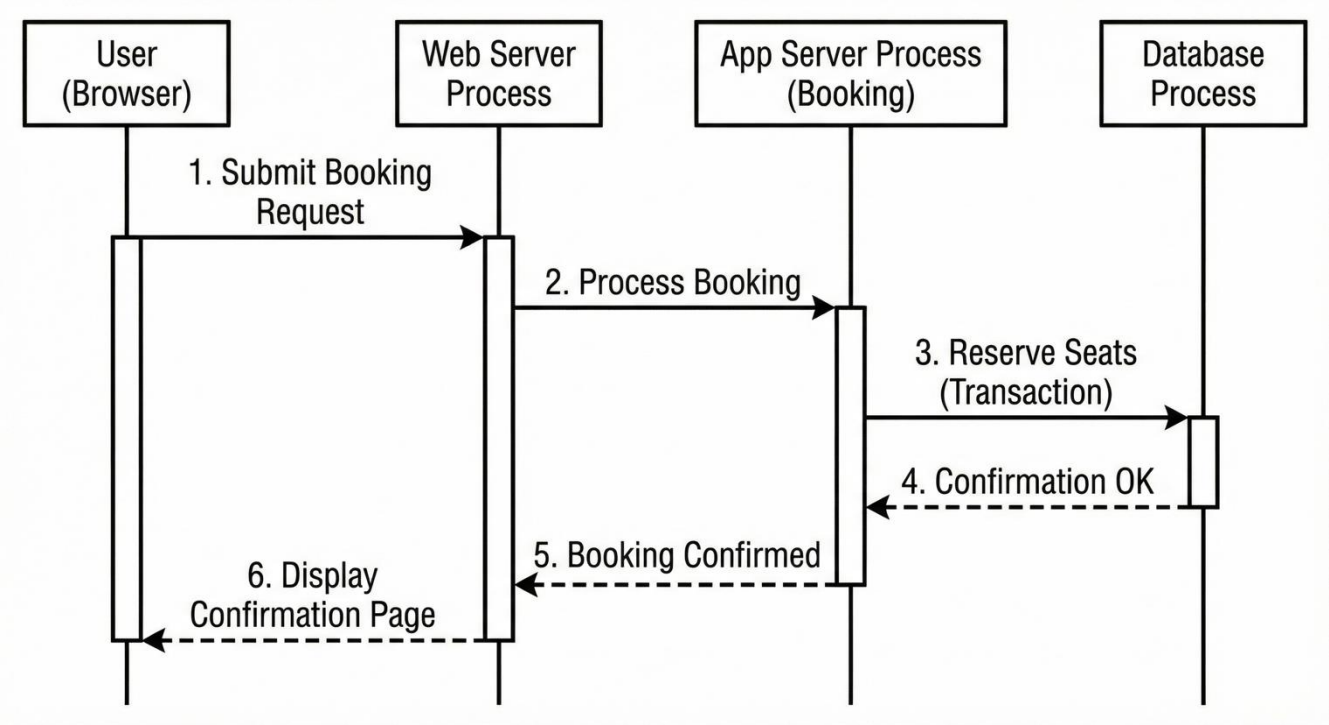
- They have a **well-defined purpose**
- Show only **necessary information**
- **Abstract away** unnecessary details
- Use legends and annotations to **remove ambiguity**

- **Multiple views** together produce a **richer understanding**

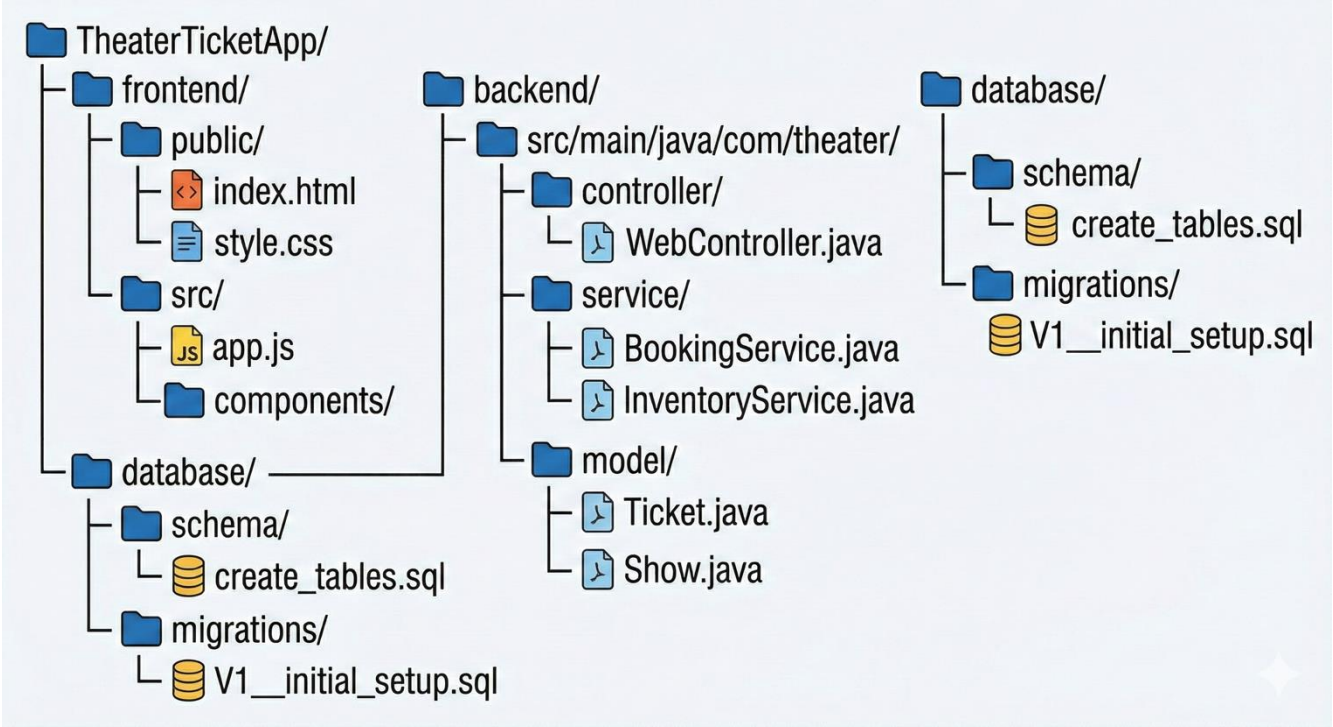
# Logical View



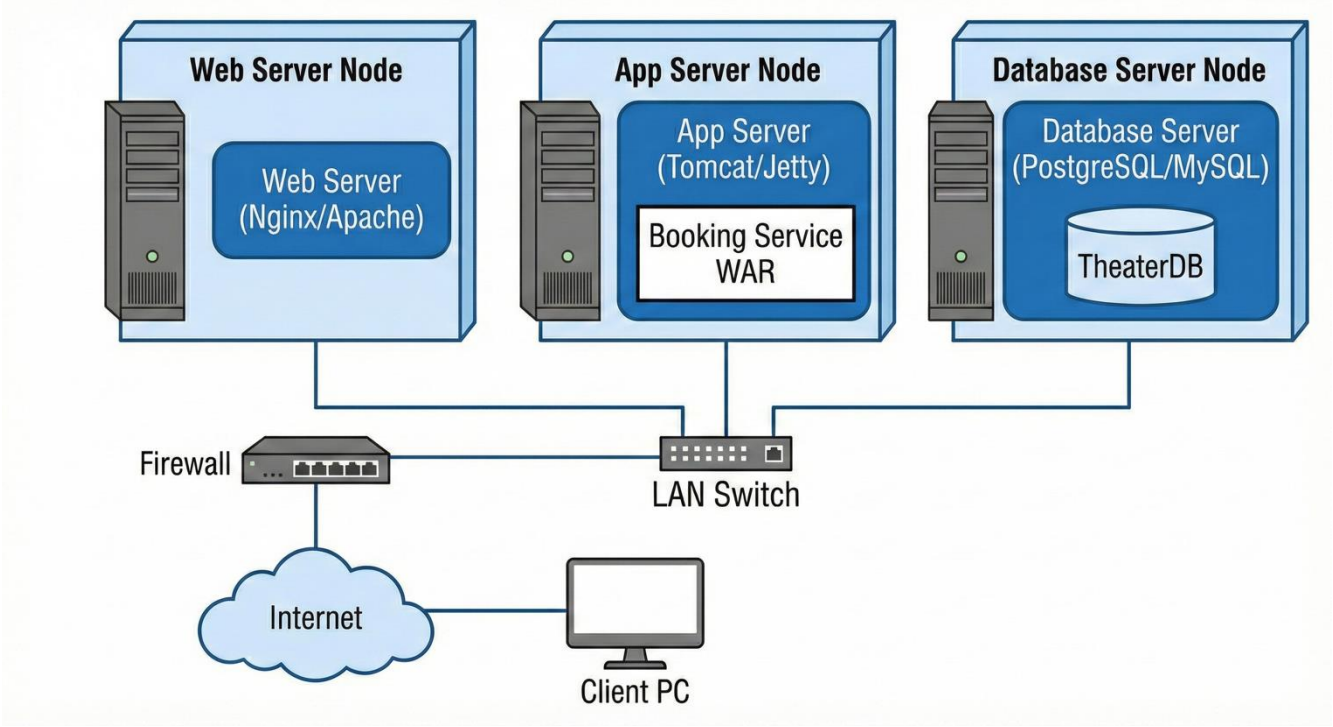
# Process View



# Development View



# Physical View



# Architecting Software



https://www.archdaily.com/

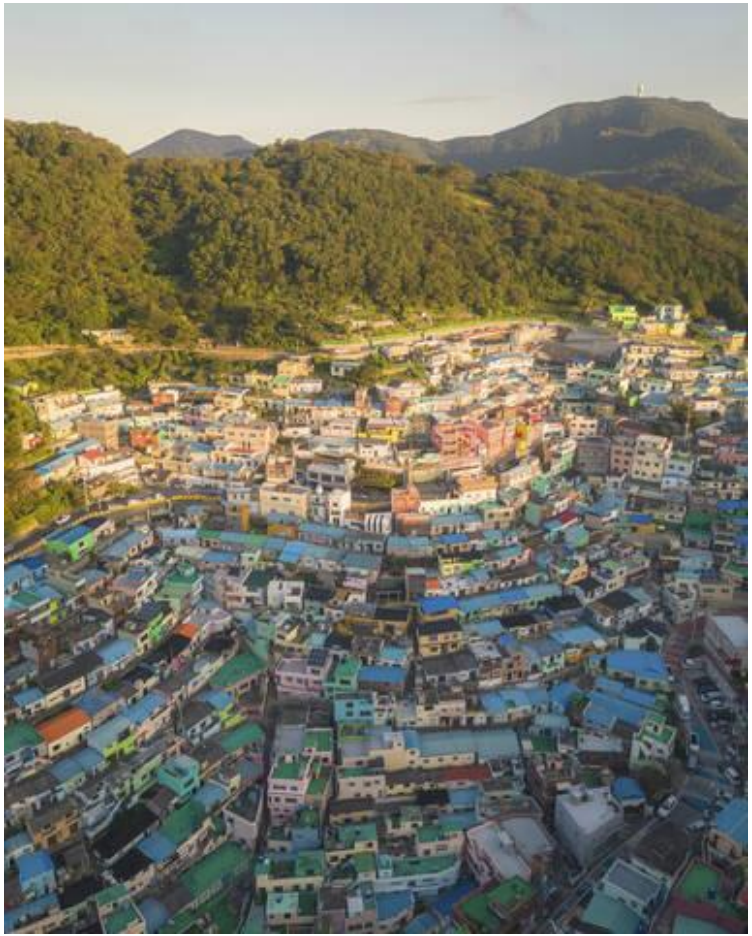


https://www.instagram.com/architectanddesign



https://www.mykonosceramica.com/





www.qr-view.com

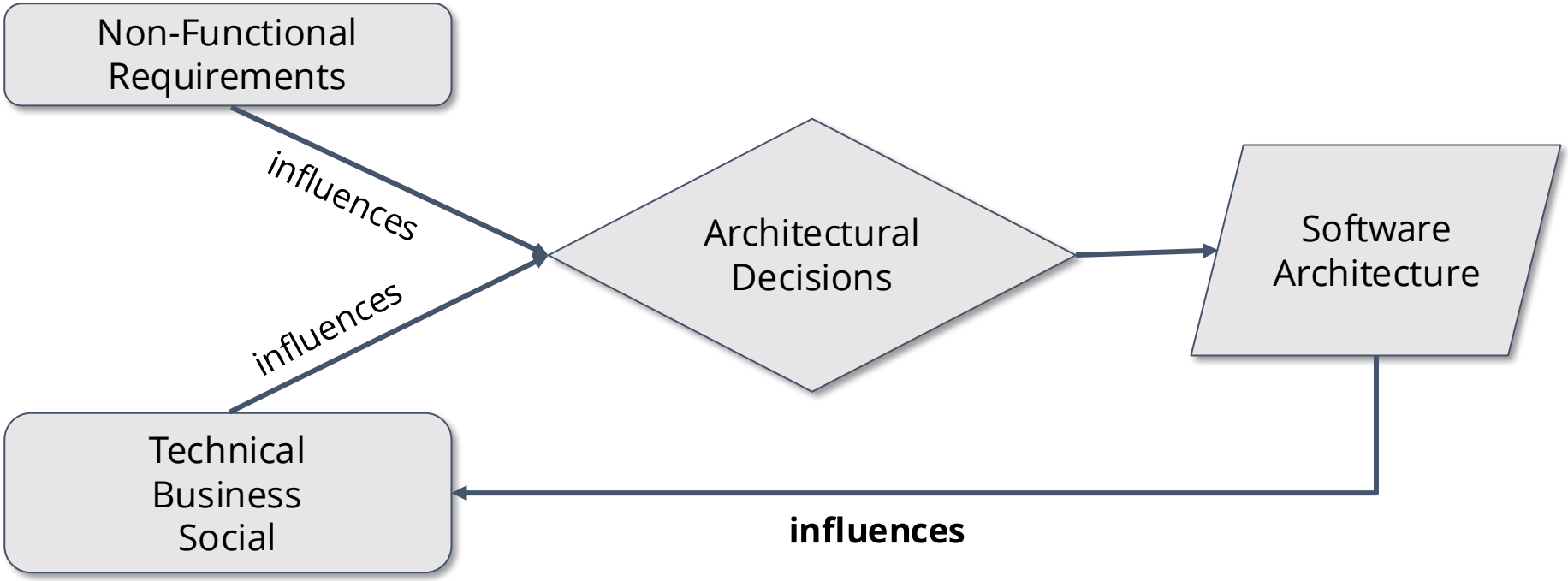
# Every system has an architecture

- Whether you know it or not
- Whether you like it or not
- Whether it is documented or not

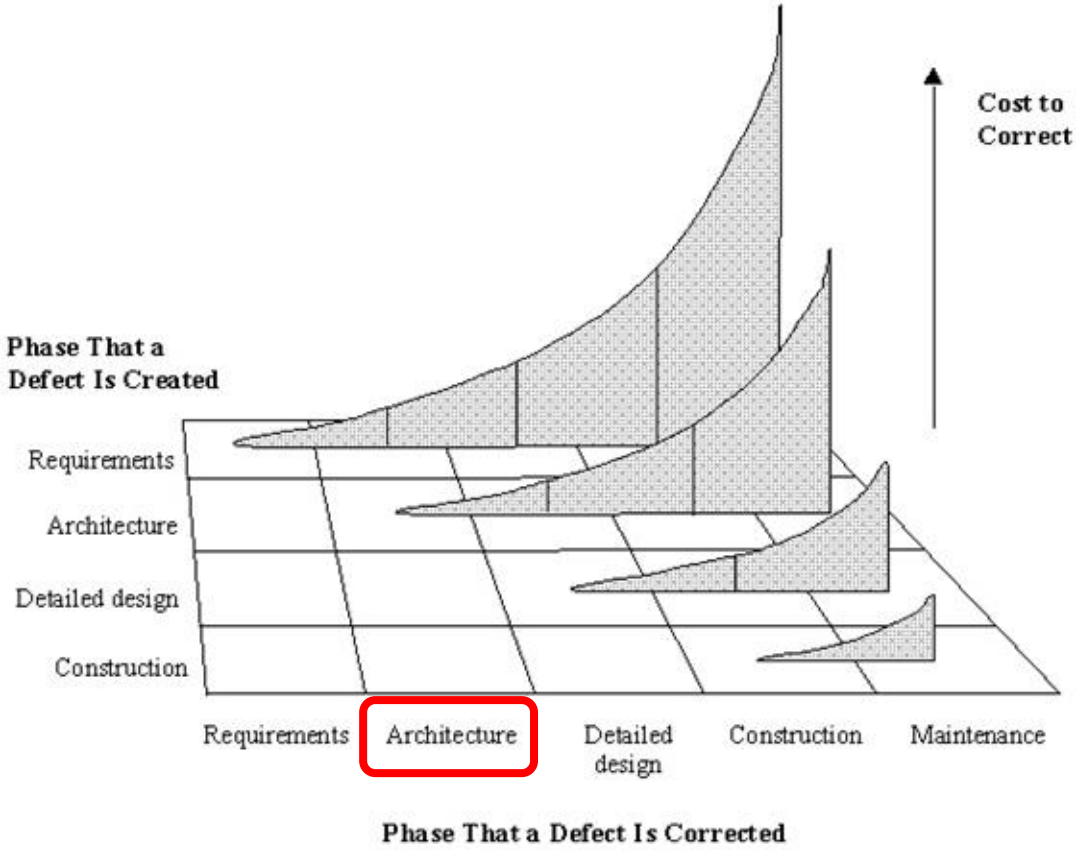
**If you don't consciously elaborate the architecture, it will evolve by itself!**




*Architecting Software the SEI Way - Software Architecture Fundamentals: Technical, Business, and Social Influences.* Robert Wojcik. 2012

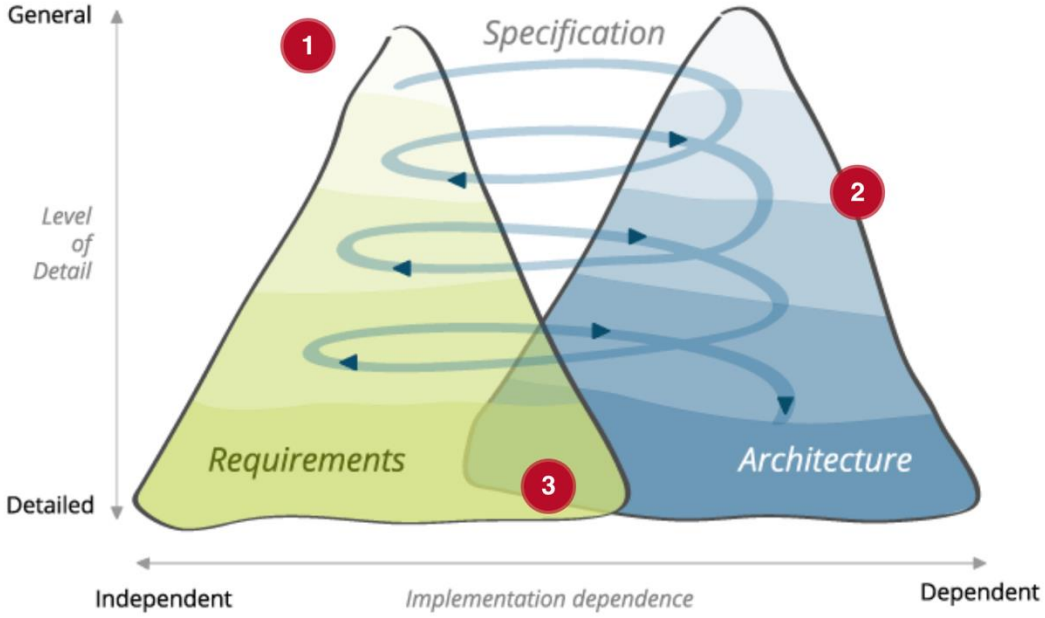



*Architecting Software the SEI Way - Software Architecture Fundamentals: Technical, Business, and Social Influences.* Robert Wojcik. 2012




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# The Twin Peaks Model




B. Nuseibeh, "Weaving together requirements and architectures". 2001

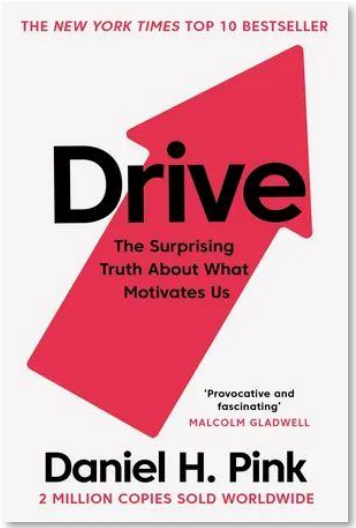
# Agile and Architecture

*"The best architectures, requirements, and designs **emerge** from **self-organizing teams**".* *The Twelve Principles of the Agile Manifesto*




*"Control leads to compliance; autonomy leads to engagement."*

Daniel H. Pink

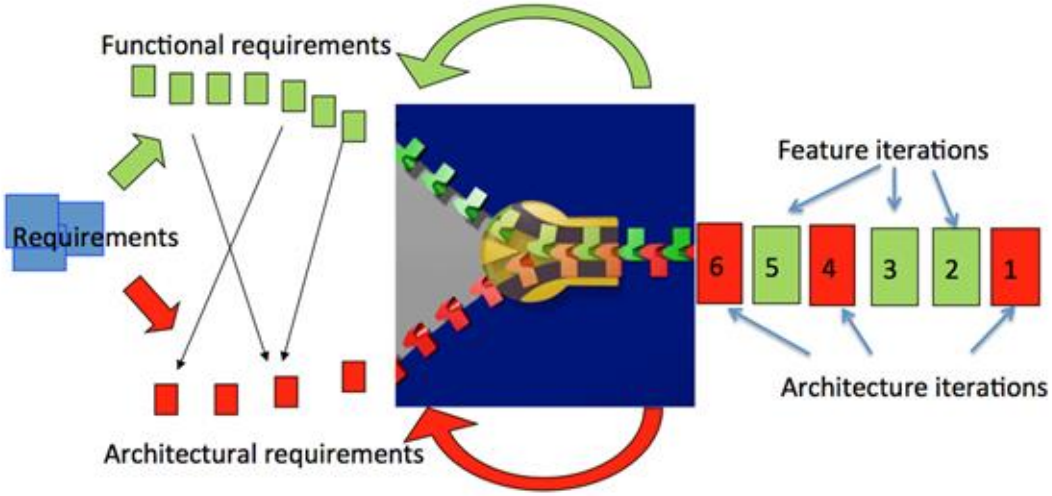



https://medium.com/@beatrix_66005/the-best-architectures-requirements-and-designs-emerge-from-self-organizing-teams-8b54ebc4c6b0

# The Zipper Model

## How to Agilely Architect an Agile Architecture

by Stephany Bellomo, Philippe Kruchten, Robert L. Nord, and Ipek Ozkaya



# Common Architectural Styles

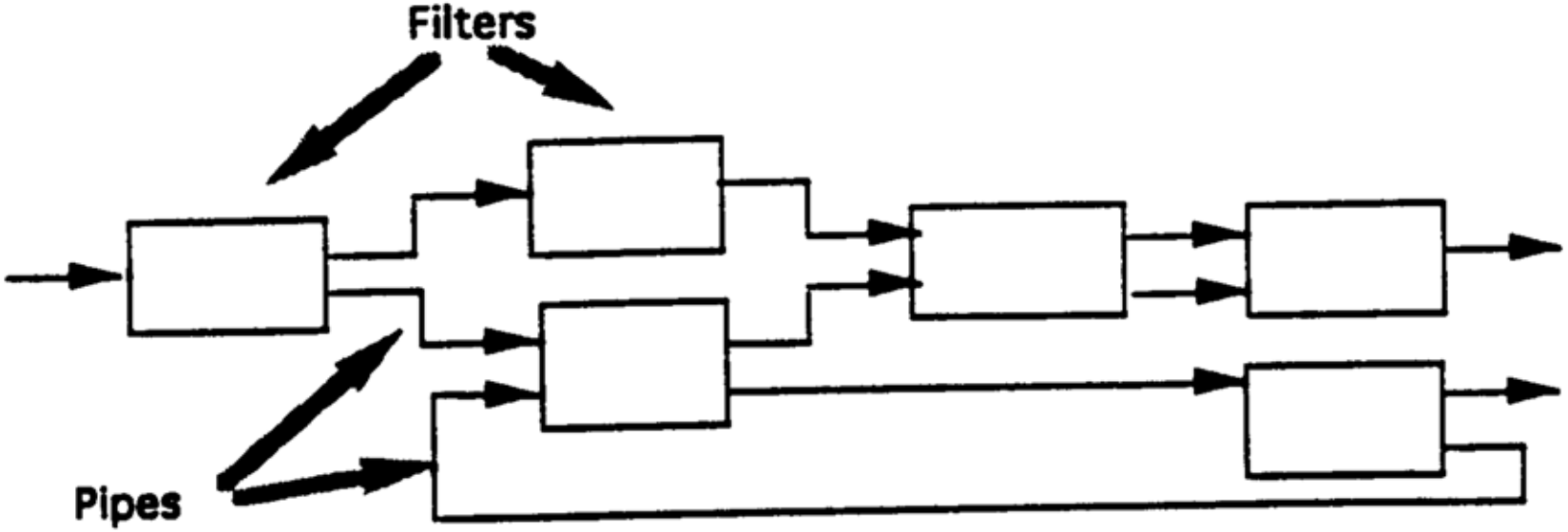






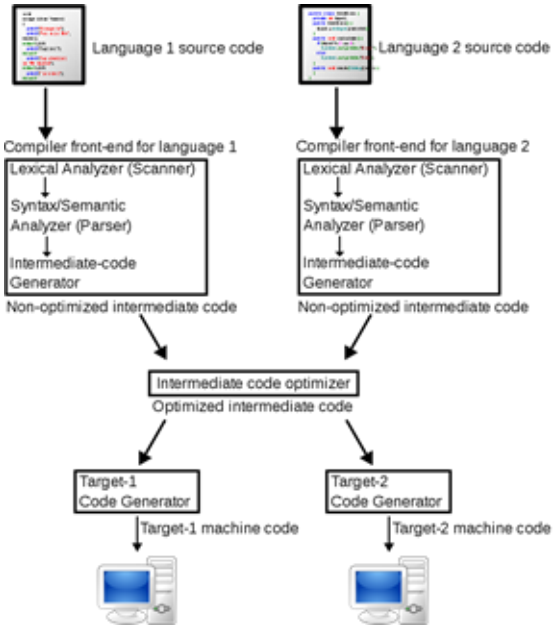

https://www.thespruce.com/top-architectural-styles-4802083

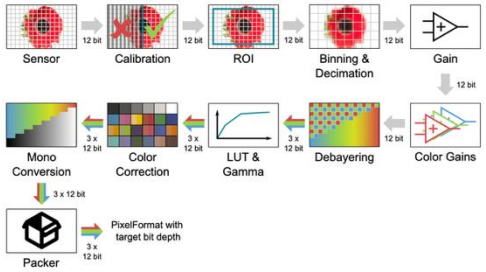
# Pipes and Filters

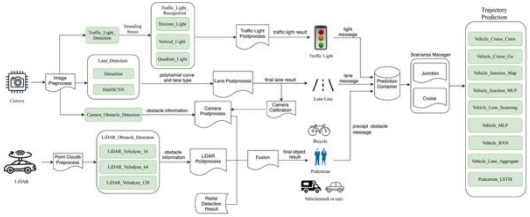


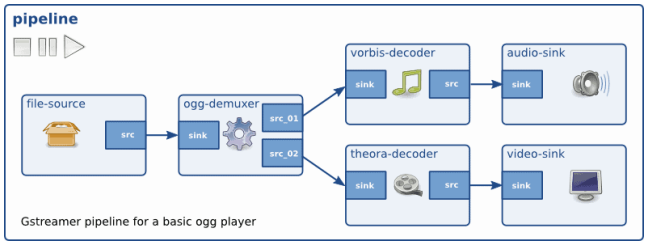

© David Garlan and Mary Shaw, CMU/SEI-94-TR-021

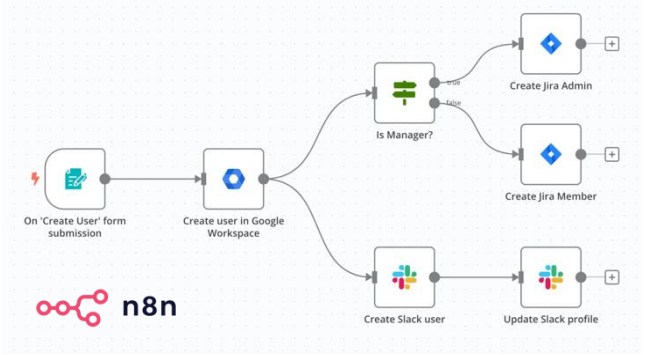
# Pipes and Filters in the Wild



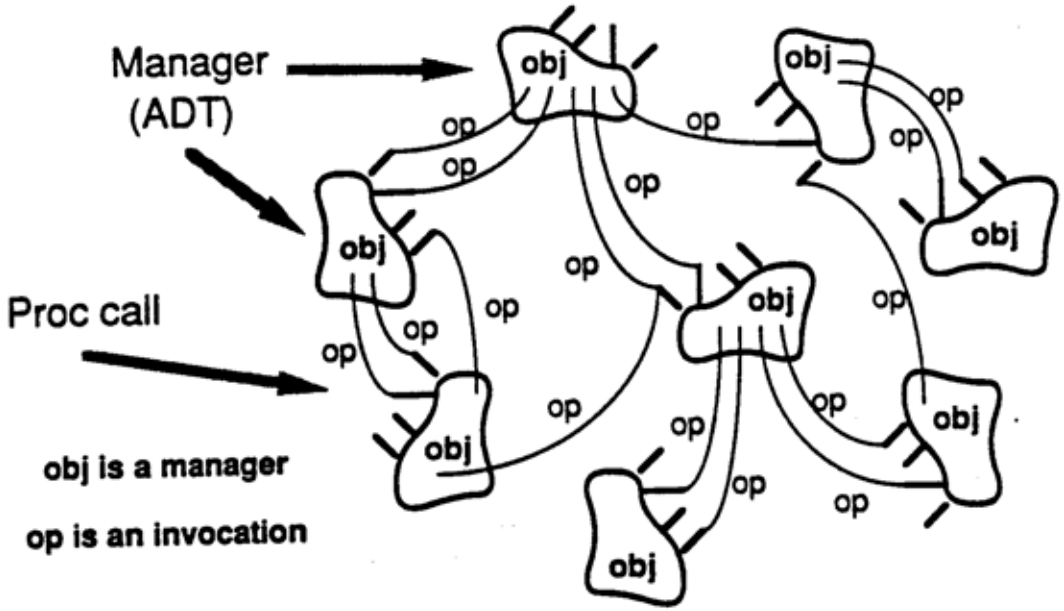





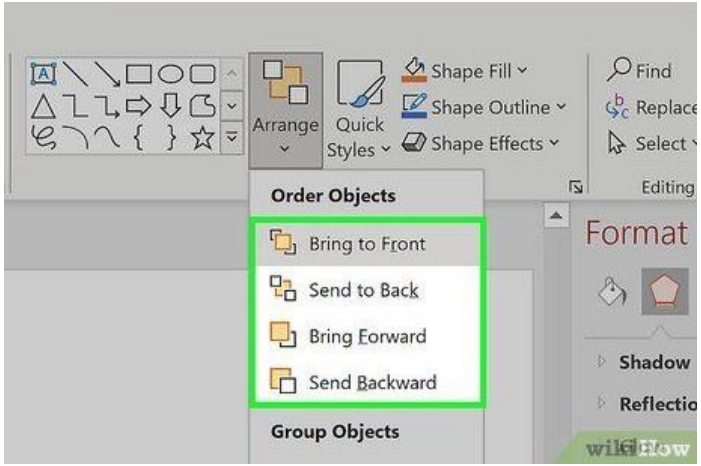



# Object-Oriented Organization



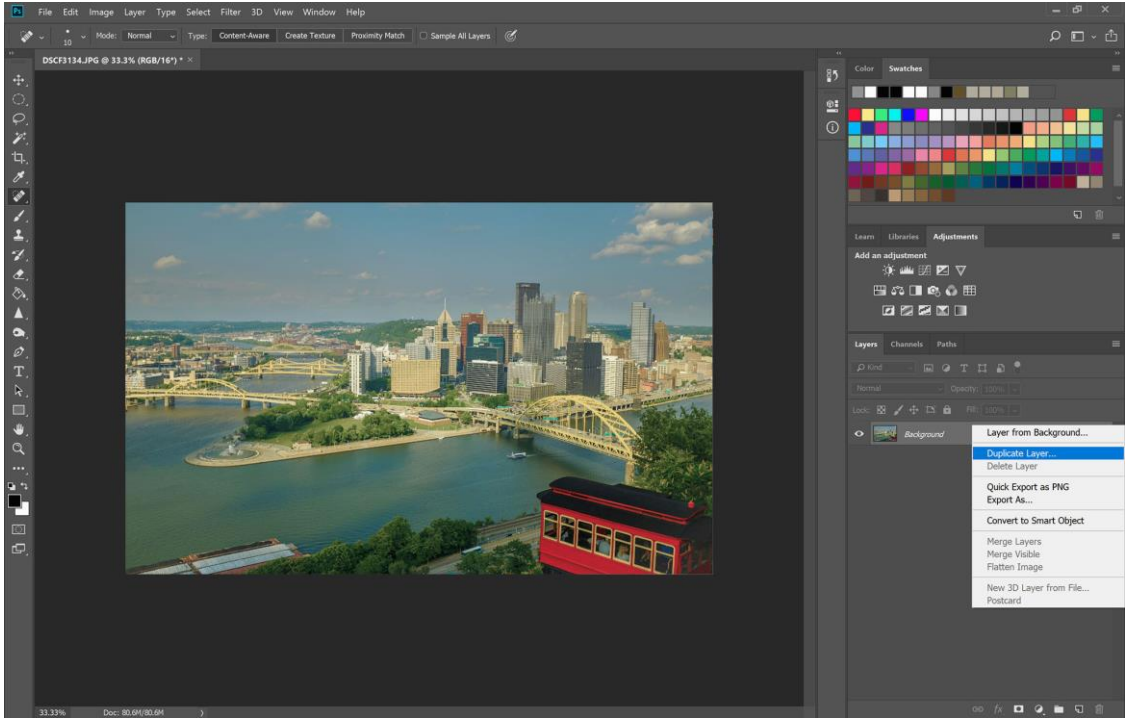

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# Object-Oriented Organization in the Wild

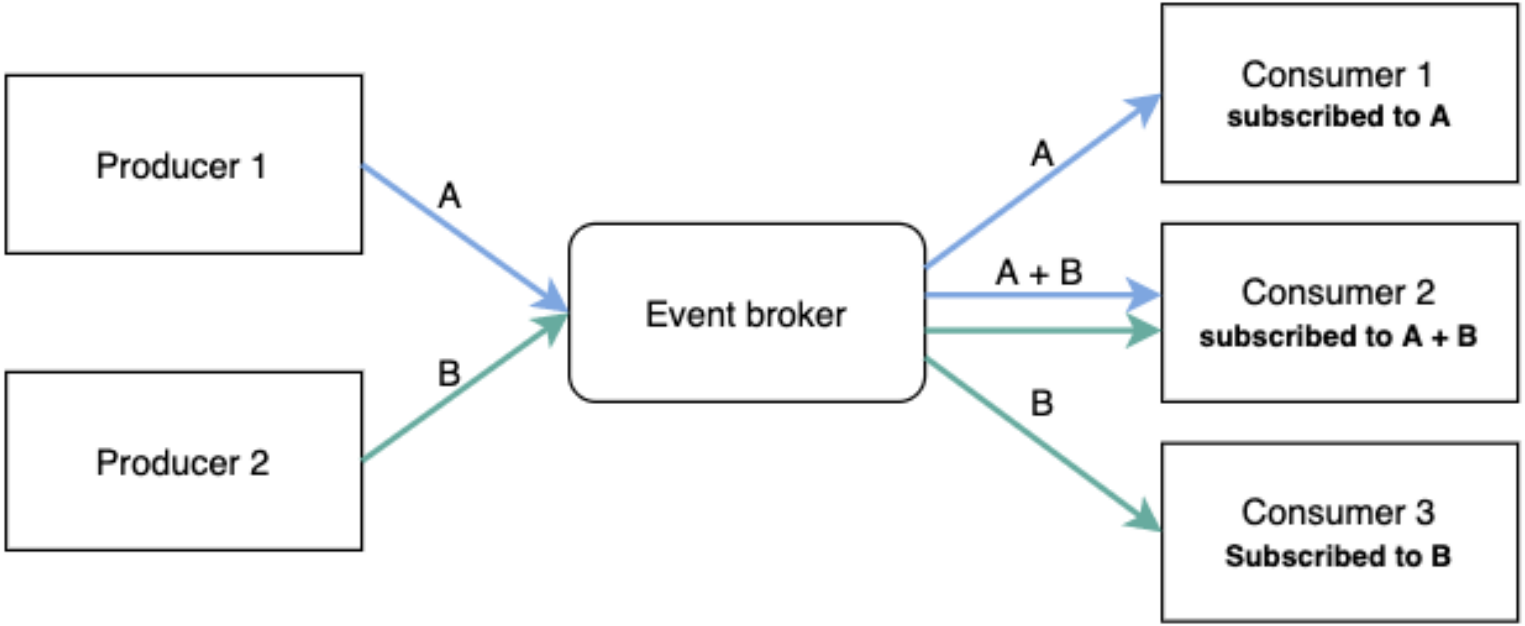




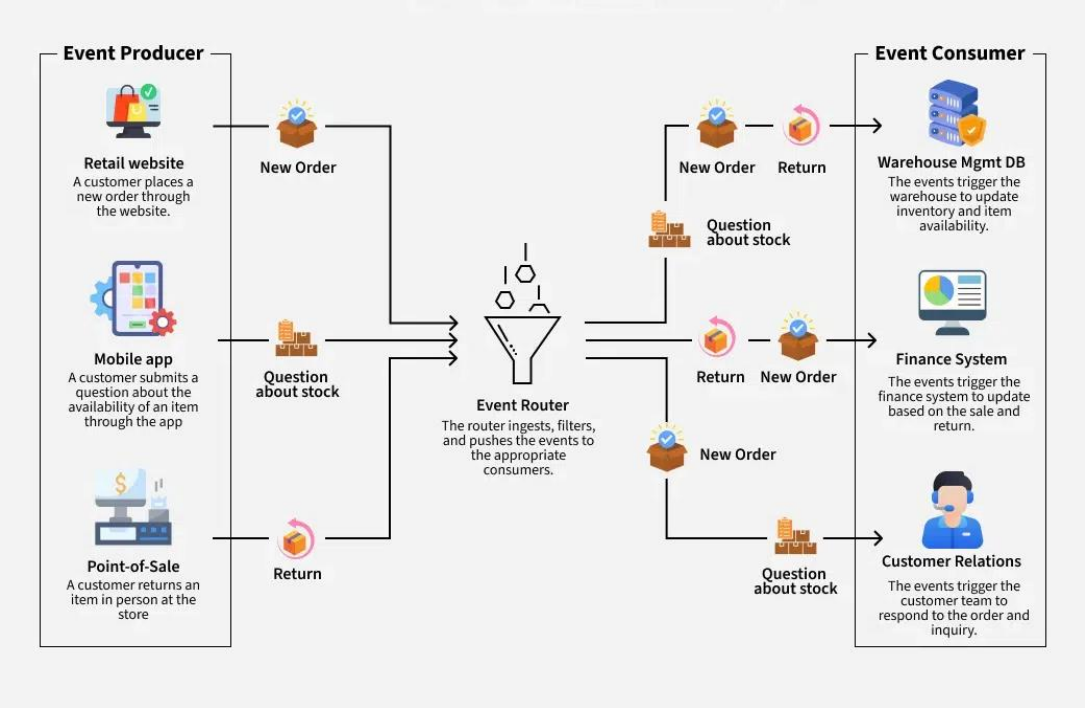


# Event-Driven Architecture



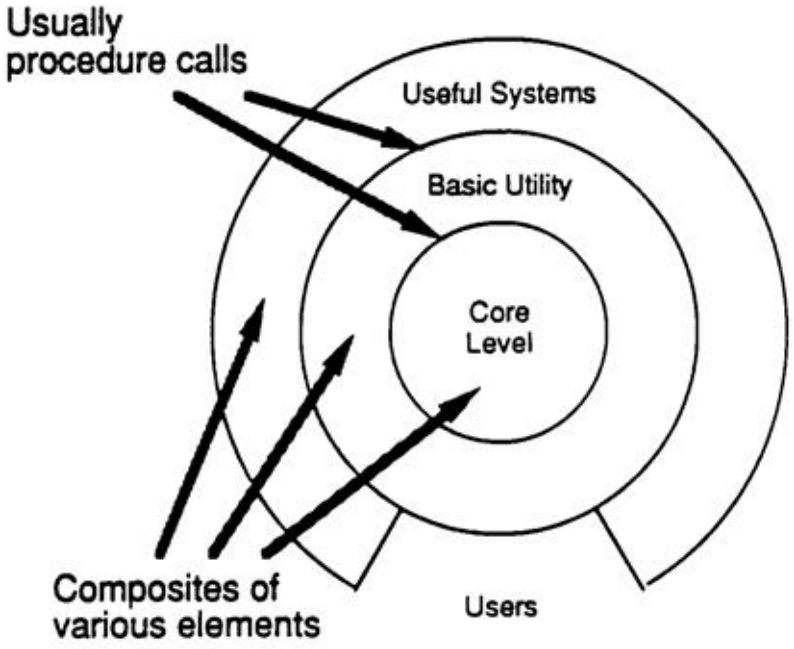
# Event-Driven Architectures in the Wild





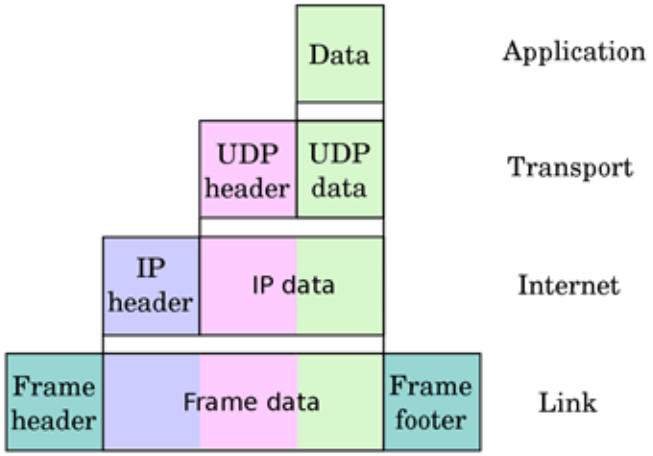


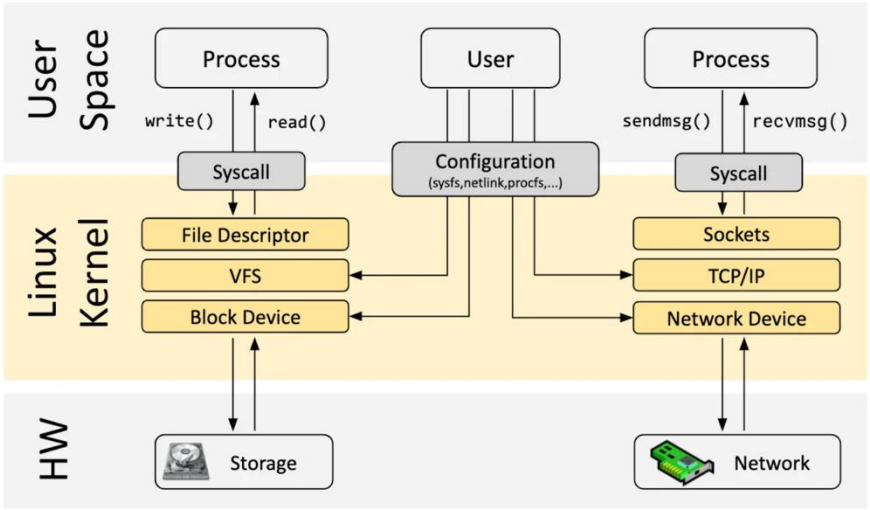

# Layered Systems




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# Layered Systems in the Wild



# Why Document Architecture?

- Blueprint for the system
  - Artifact for early analysis
  - Primary carrier of quality attributes
  - Key to post-deployment maintenance and enhancement
- Documentation speaks for the architect, both today and 20 years from today
  - As long as the system is built, maintained, and evolved according to its documented architecture
- Support traceability

**Elon Musk** ✓
@elonmusk

Btw, I'd like to apologize for Twitter being super slow in many countries.
App is doing >1000 poorly batched RPCs just to render a home timeline!

1:00 PM · Nov 13, 2022

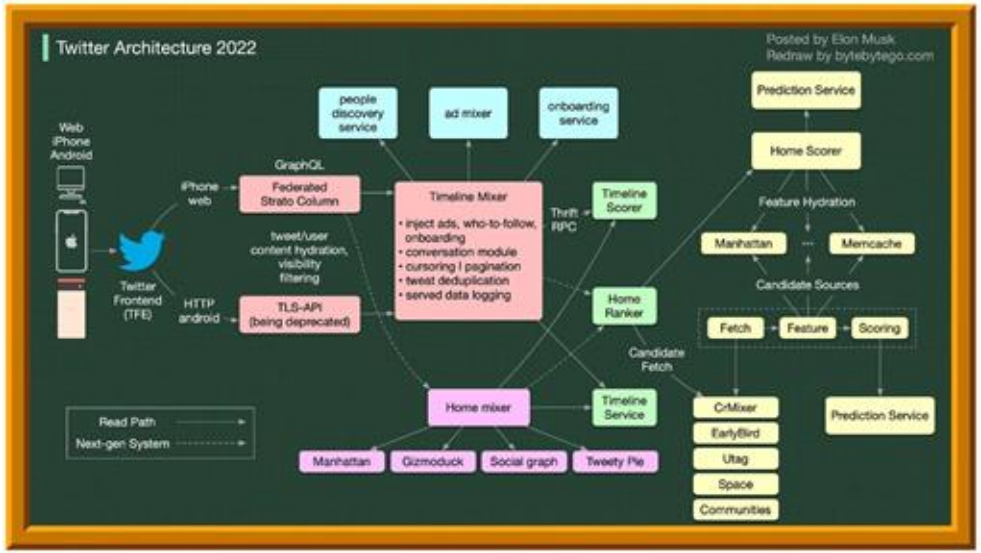
**Elon Musk** ✓
@elonmusk

Just leaving Twitter HQ code review



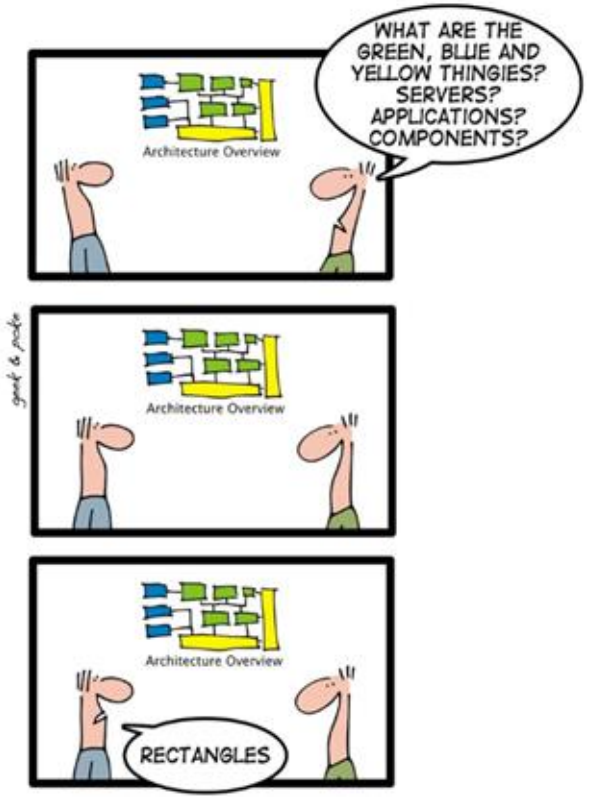
4:28 AM · Nov 19, 2022

**36.9K** Retweets **16.1K** Quote Tweets **464K** Likes



# Guidelines for selecting a notation

- Suitable for purpose
- Often visual for compact representation
- Usually, boxes and arrows
- UML possible (semi-formal), but possibly constraining
  - Note the different abstraction level – Subsystems or processes, not classes or objects
- Formal notations available
- Decompose diagrams hierarchically and in views
- Always include a legend
- Define precisely what the boxes mean
- Define precisely what the lines mean
- Do not try to do too much in one diagram
  - Each view of architecture should fit on a page
  - Use hierarchy

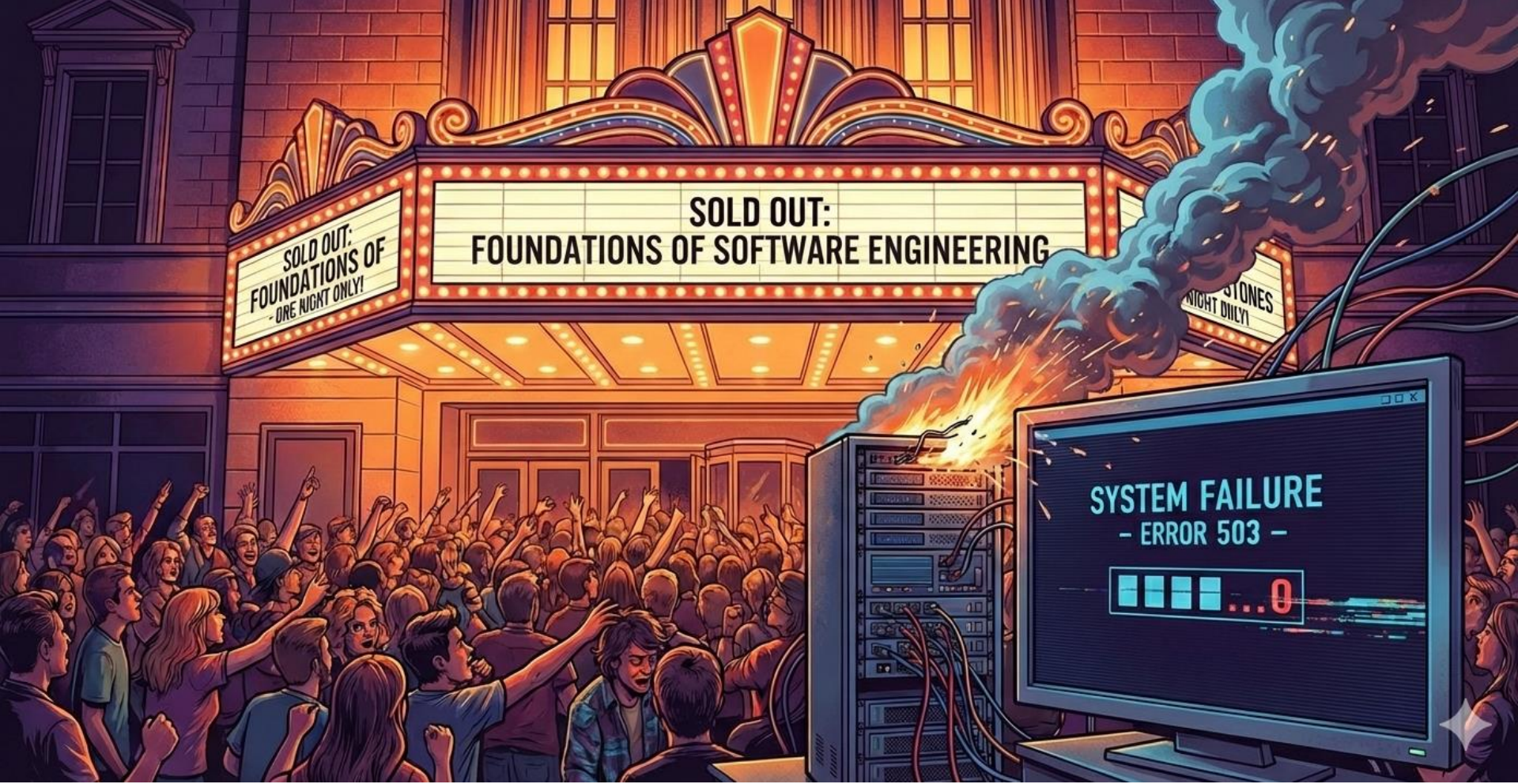


# Case Study: Victim of Success Theater

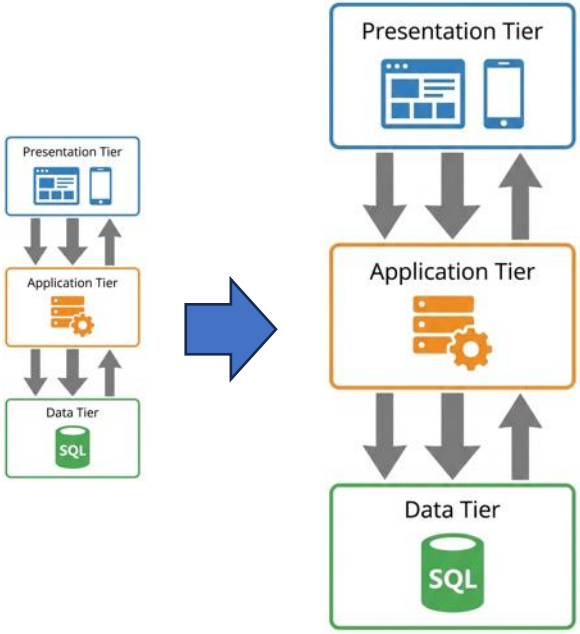
# Case Study: Victim of Success Theater

- Simple web application for a local theater:
  - Sell tickets
  - 500 seats
- Architecture: Standard "3-Tier" setup:
  UI → API → Database
- Happy path :
  - A user clicks "Buy”
  - API checks the DB, decrements the ticket count, and sends a confirmation
  - It works perfectly for months.

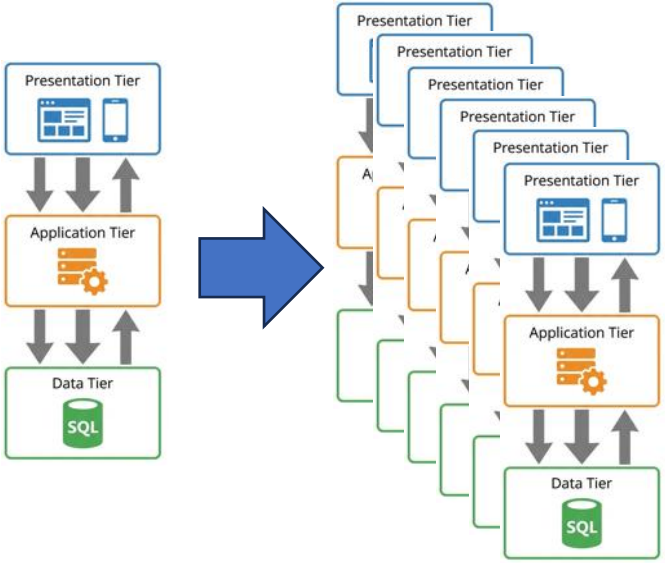



SOLD OUT:
FOUNDATIONS OF SOFTWARE ENGINEERING
SOLD OUT:
FOUNDATIONS OF
- ONE NIGHT ONLY!
SYSTEM FAILURE
- ERROR 503 -

# How can we scale to 1000 theaters?




OR?

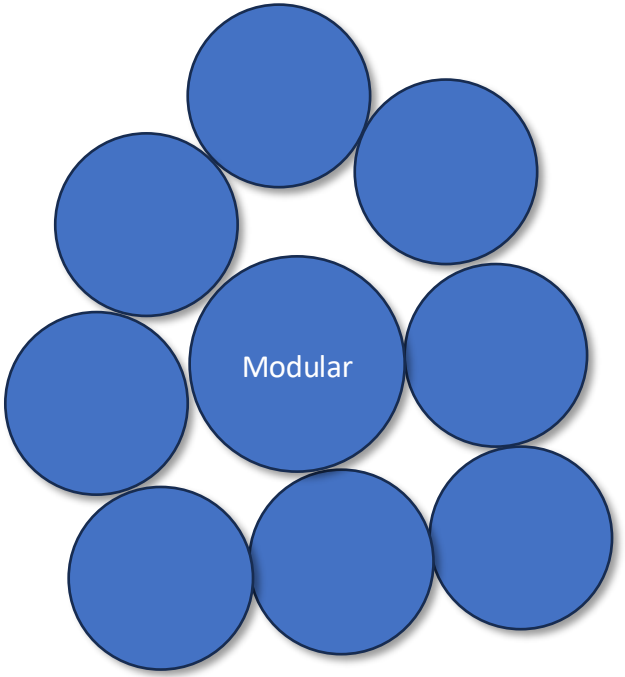



Vertical Scaling

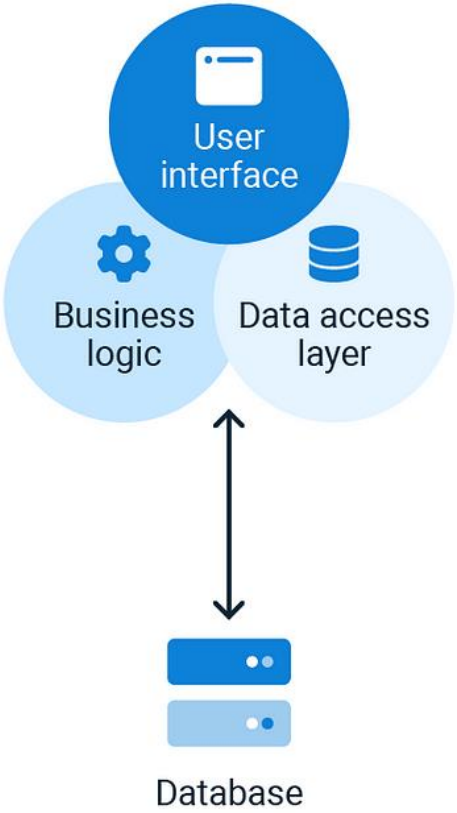
Horizontal Scaling

# Monolithic vs. Modular Architectures

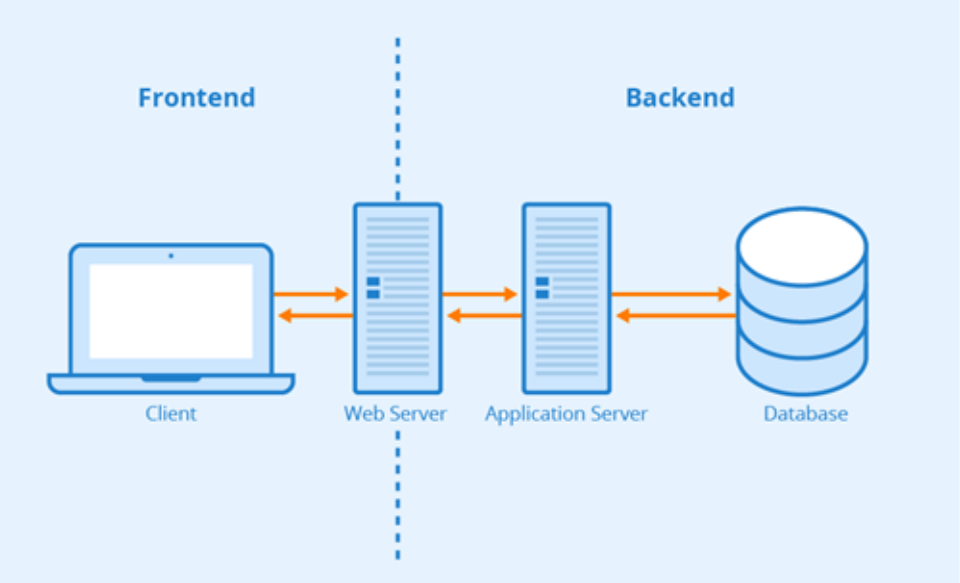

Monolithic

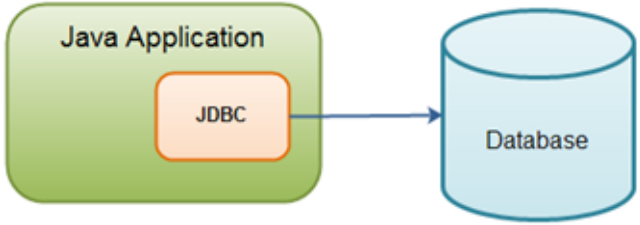

Modular

# Monolithic Architecture



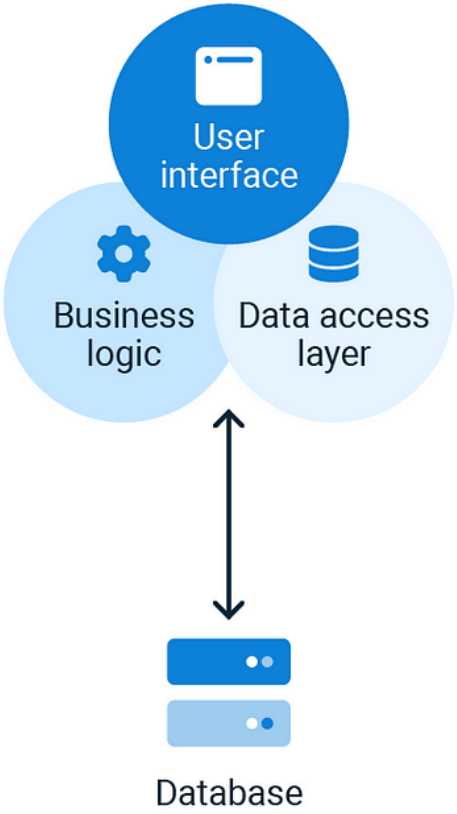
# Monolithic Styles



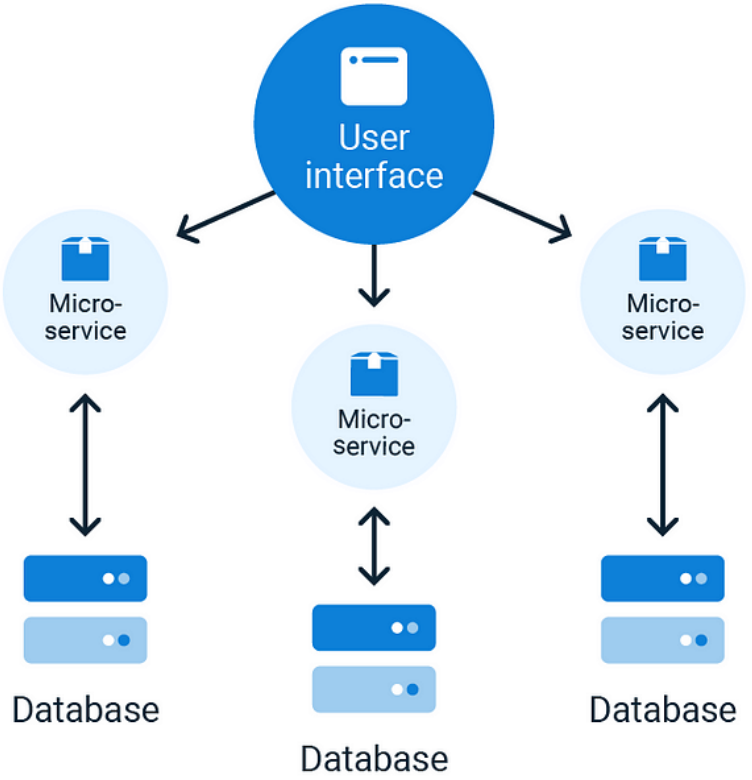




Source: https://www.seobility.net (CC BY-SA 4.0)

## Monolithic Architecture




## Microservice Architecture



# MICROSERVICES


Microservices
Microservices Everywhere

# Microservices













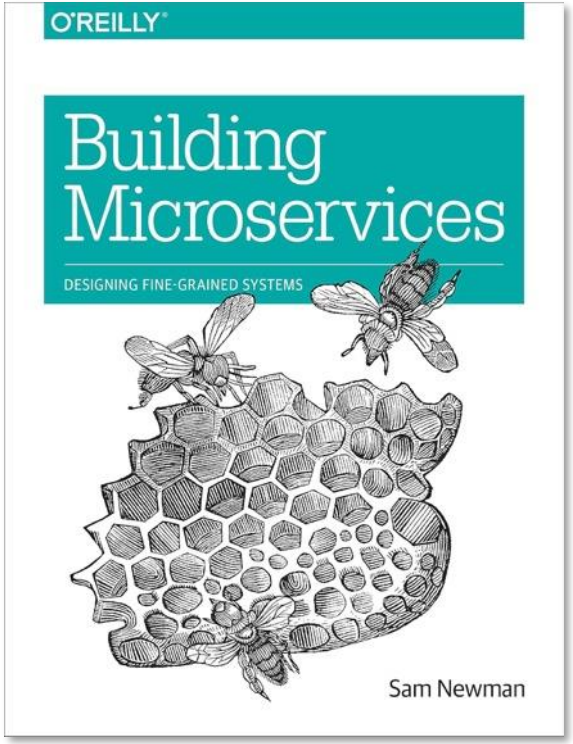






"*Small **autonomous** services that work well together*"

Sam Newman



# Monolith vs. Service-Based vs. Microservice



# Netflix Microservices



# Why Can't Netflix Use a Monolithic Architecture?

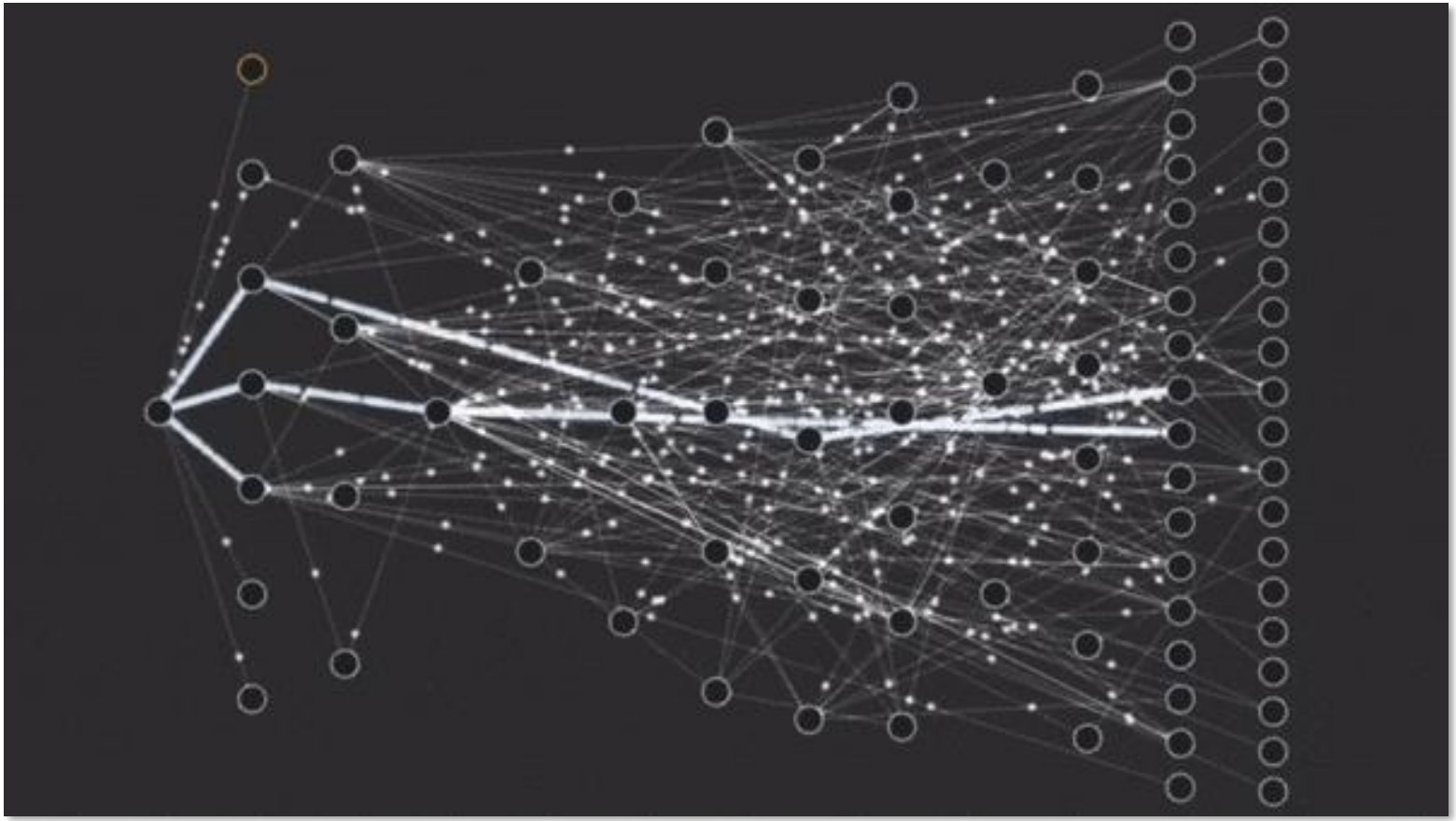


- Requires an architecture that can handle **various computational demands**
- Need scalability must support **millions of users worldwide**
- Need fault tolerance to maintain a **seamless user experience**
- New features and improvements need to be **rolled out rapidly**

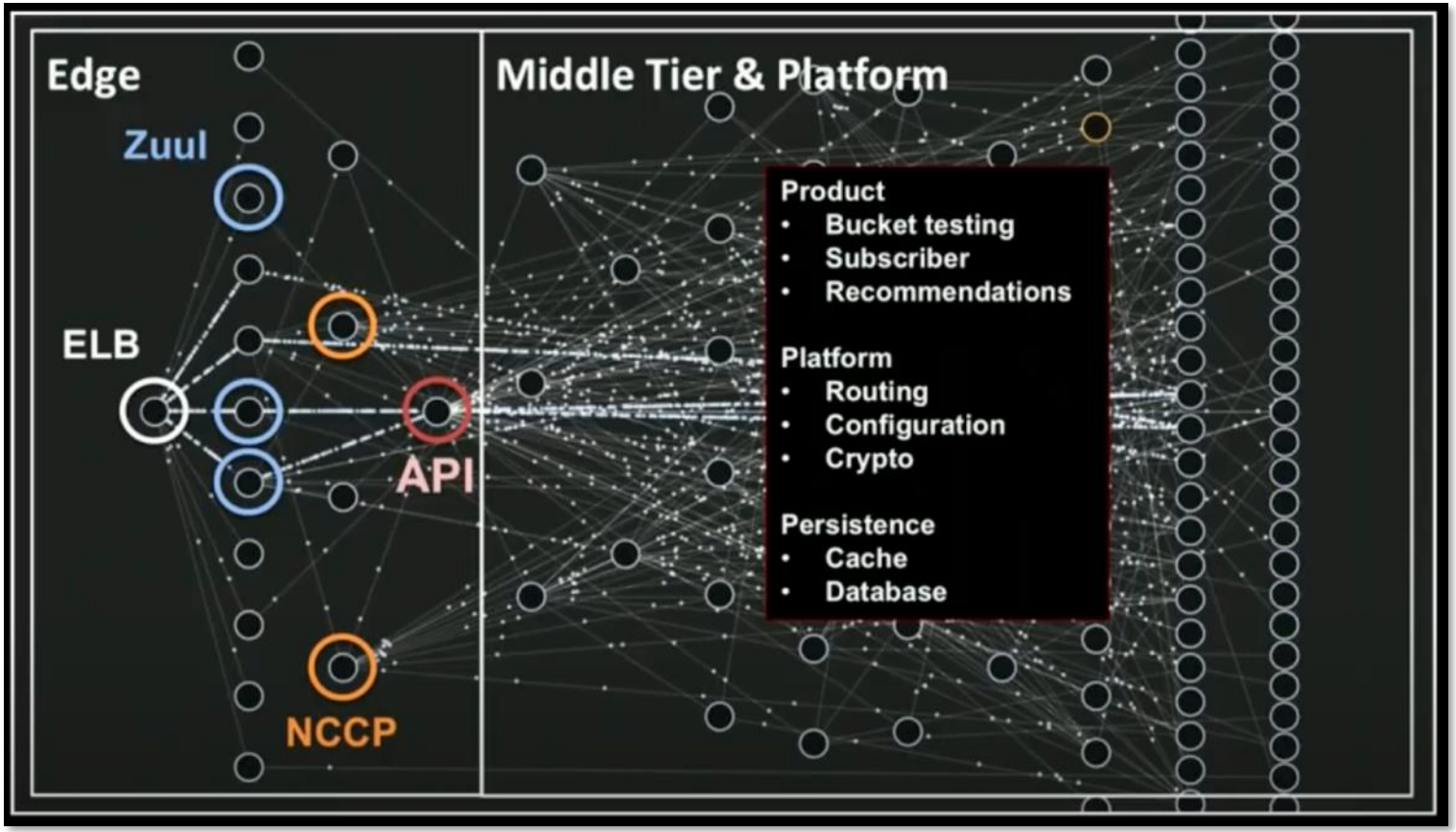
# Netflix Microservices




- User Subscriptions
- Banner Ad
- Popular Shows
- Trending Now
- Continue Watching
- My List (saved shows)
- Notifications
- User Management
- Subtitles
- …

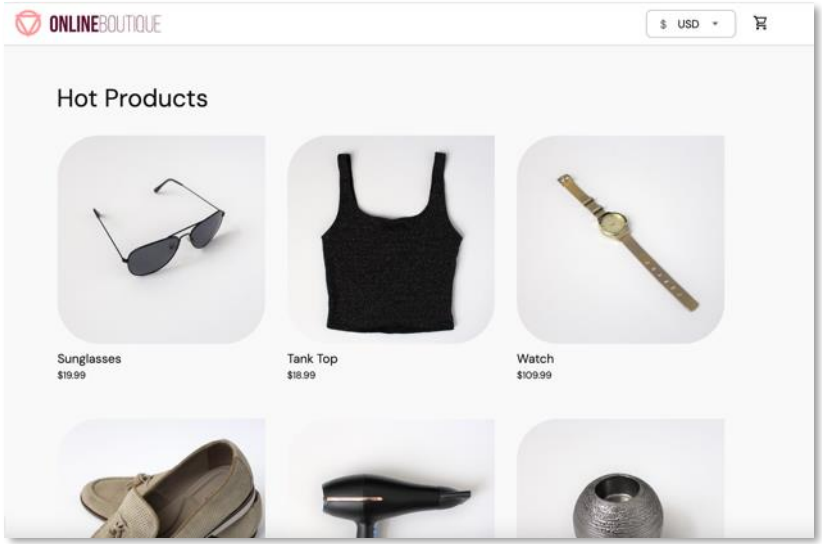


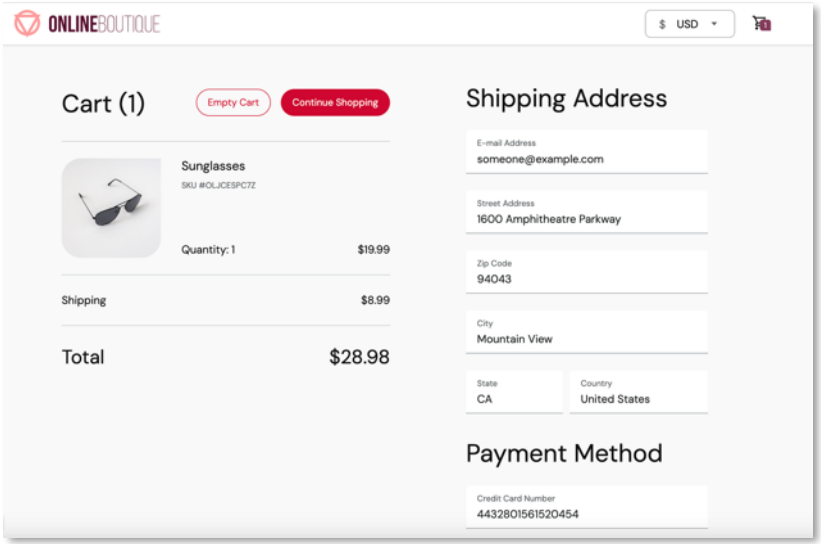
https://www.youtube.com/watch?v=V_oxbj-a1wQ




https://www.youtube.com/watch?v=V_oxbj-a1wQ

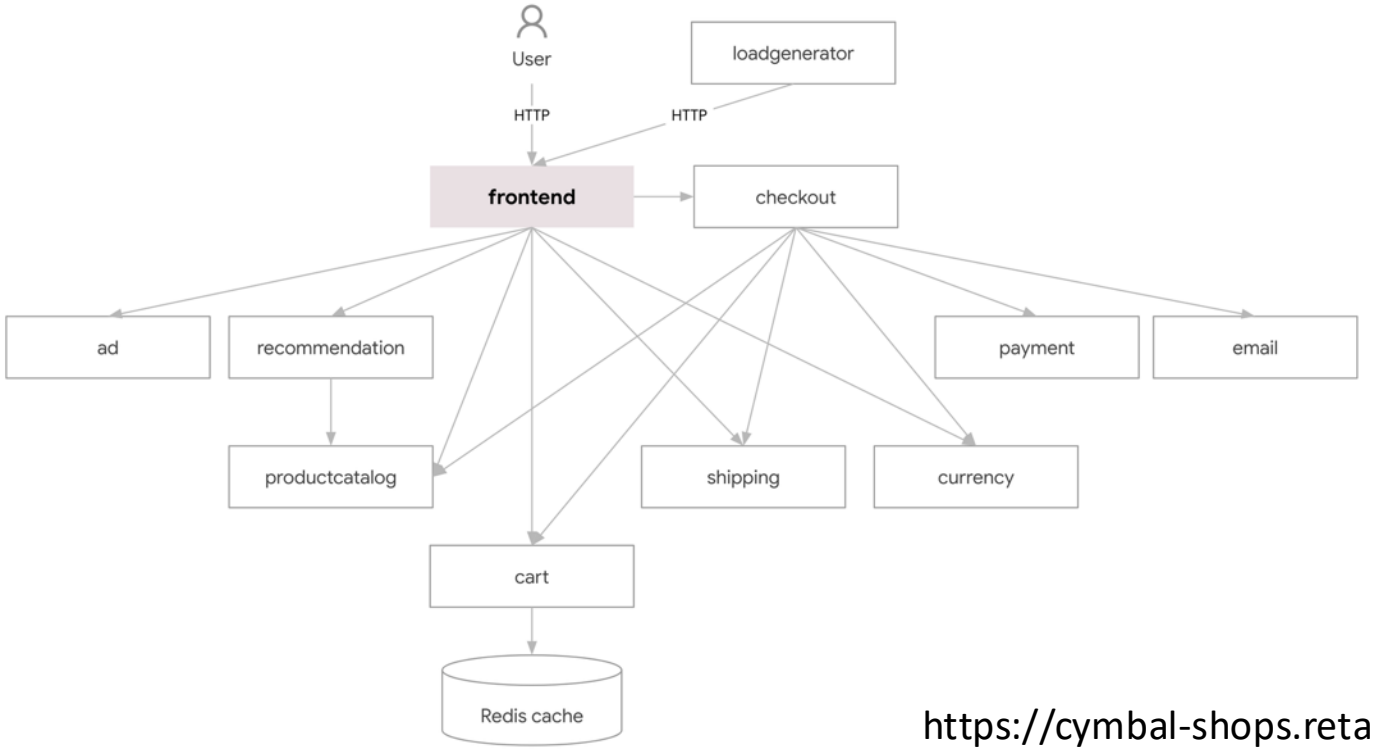
# Online Boutique: Guess some microservices




https://cymbal-shops.retail.cymbal.dev

# Online Boutique: Microservice Architecture




https://cymbal-shops.retail.cymbal.dev

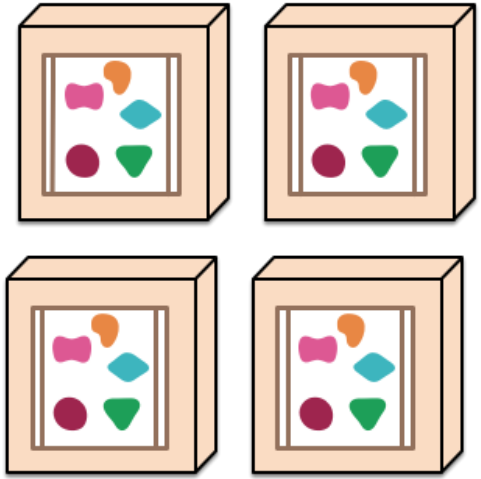
| Service | Language | Description |
|---|---|---|
| frontend | Go | Exposes an HTTP server to serve the website. Does not require signup/login and generates session IDs for all users automatically. |
| cartservice | C# | Stores the items in the user's shopping cart in Redis and retrieves it. |
| productcatalogservice | Go | Provides the list of products from a JSON file and ability to search products and get individual products. |
| currencyservice | Node.js | Converts one money amount to another currency. Uses real values fetched from European Central Bank. It's the highest QPS service. |
| paymentservice | Node.js | Charges the given credit card info (mock) with the given amount and returns a transaction ID. |
| shippingservice | Go | Gives shipping cost estimates based on the shopping cart. Ships items to the given address (mock) |
| emailservice | Python | Sends users an order confirmation email (mock). |
| checkoutservice | Go | Retrieves user cart, prepares order and orchestrates the payment, shipping and the email notification. |
| recommendationservice | Python | Recommends other products based on what's given in the cart. |
| adservice | Java | Provides text ads based on given context words. |
| loadgenerator | Python/Locust | Continuously sends requests imitating realistic user shopping flows to the frontend. |

# Scalability

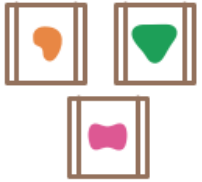
*A monolithic application puts all its functionality into a single process...*



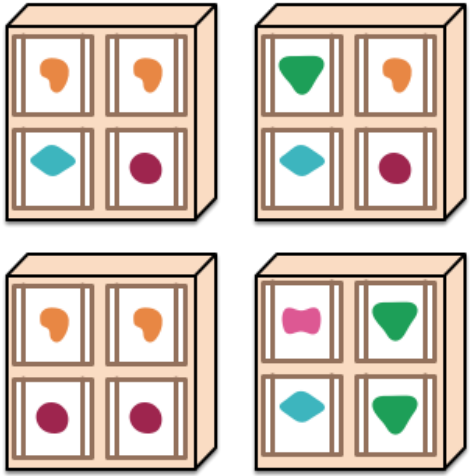
*... and scales by replicating the monolith on multiple servers*



*A microservices architecture puts each element of functionality into a separate service...*



*... and scales by distributing these services across servers, replicating as needed.*

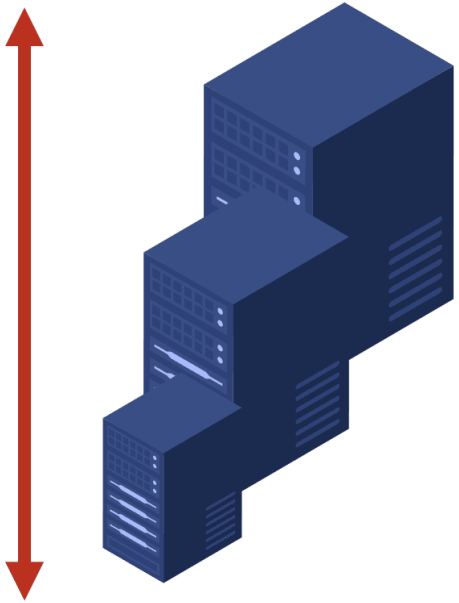


Source: http://martinfowler.com/articles/microservices.html

# Types of Scaling: Vertical vs. Horizontal

## Vertical Scaling

Increase or decrease the capacity of existing services/instances.

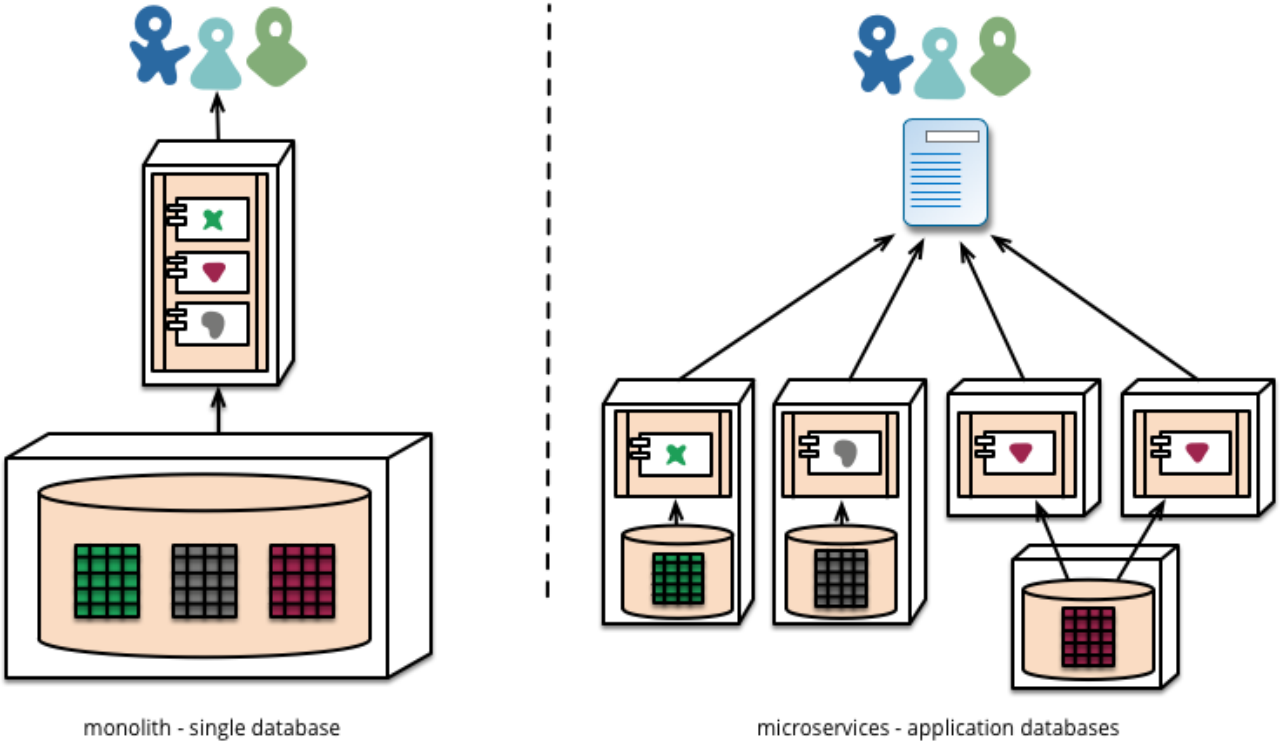


## Horizontal Scaling

Add more resources like virtual machines to your system to spread out the workload across them.



# Data Management and Consistency



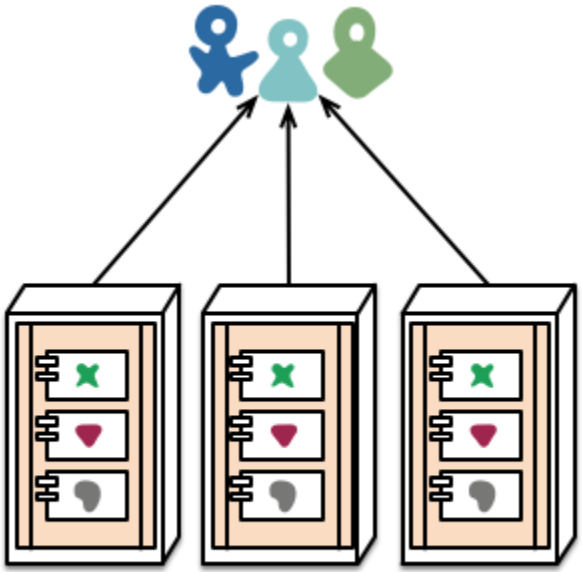
monolith - single database

microservices - application databases

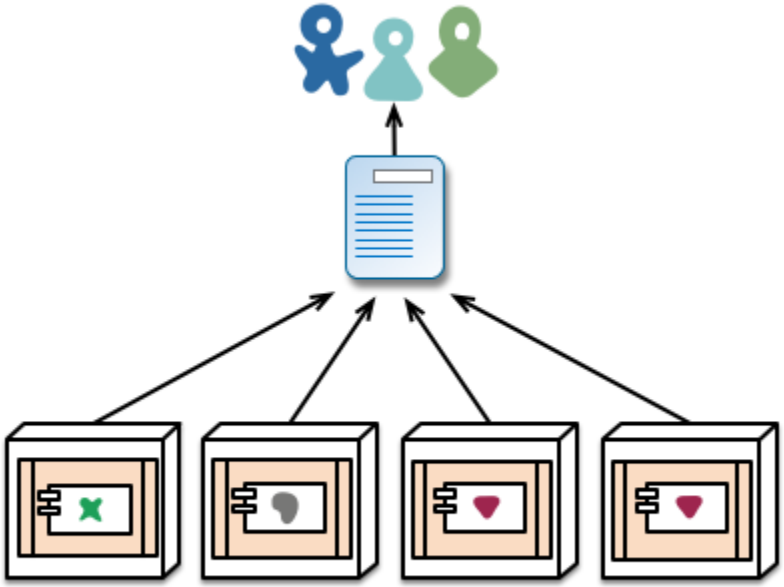


Source: http://martinfowler.com/articles/microservices.html

# Deployment and Evolution



monolith - multiple modules in the same process



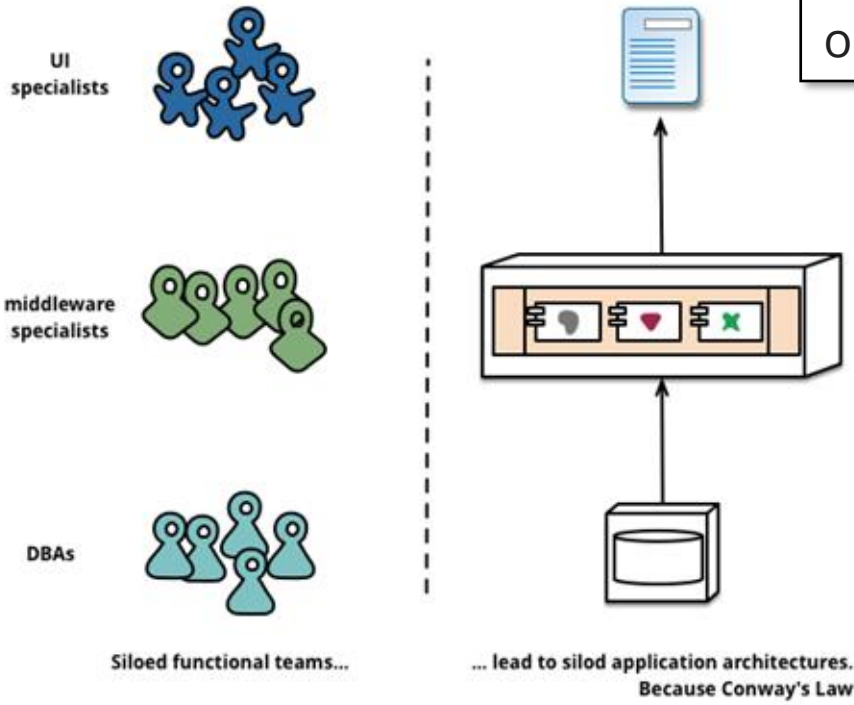
microservices - modules running in different processes

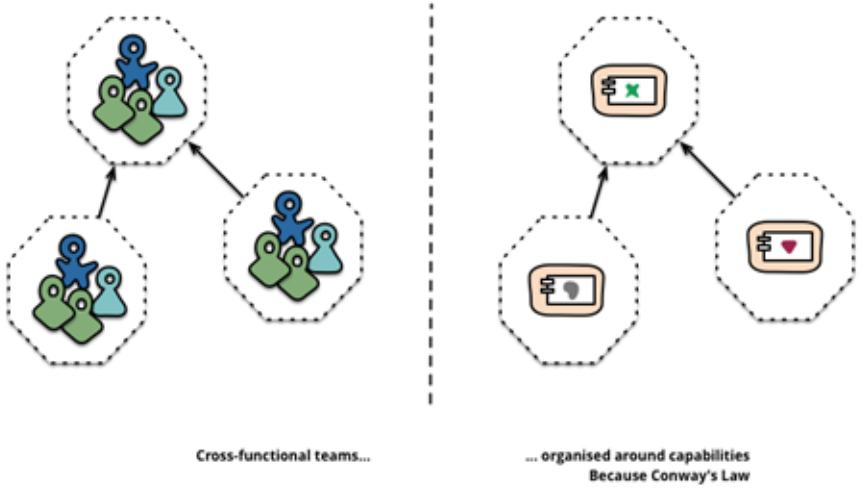


Source: http://martinfowler.com/articles/microservices.html

# Conway's Law

"Any organization that designs a system (defined broadly) will produce a design whose structure is a copy of the organization's communication structure."




"Products" not "Projects"

# YOU BUILD IT
# YOU RUN
# ~~AWAY~~ IT

"The traditional model is that you take your software to the wall that separates development and operations, and throw it over and then forget about it. Not at Amazon. You build it, you run it. This brings developers into contact with the day-to-day operation of their software. It also brings them into day-to-day contact with the customer. This customer feedback loop is essential for improving the quality of the service."

-- Werner Vogels in "A conversation with Werner Vogels" in ACM Queue, May 2006

# MICROSERVICES: PRINCIPLES

Domain Driven Modeling

Culture of Automation

Hide Implementation Details

Decentralized Governance

Deploy Independently

Consumer First

Isolate Failures

Sam Newman's Principles of Microservices

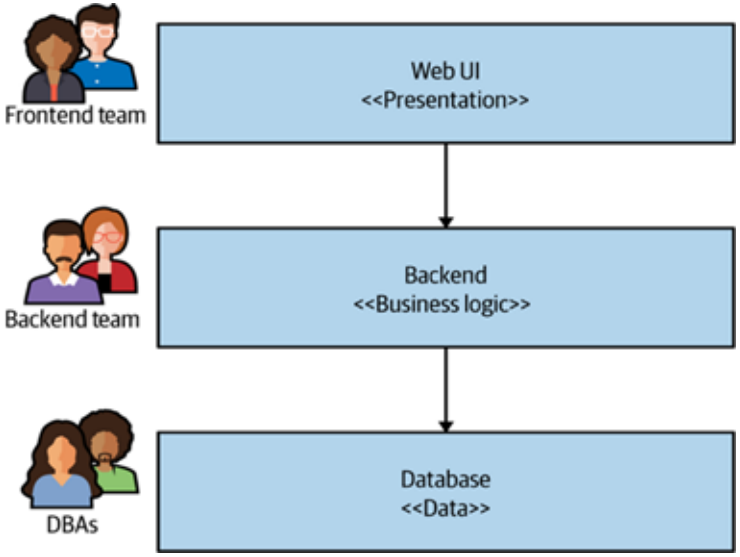


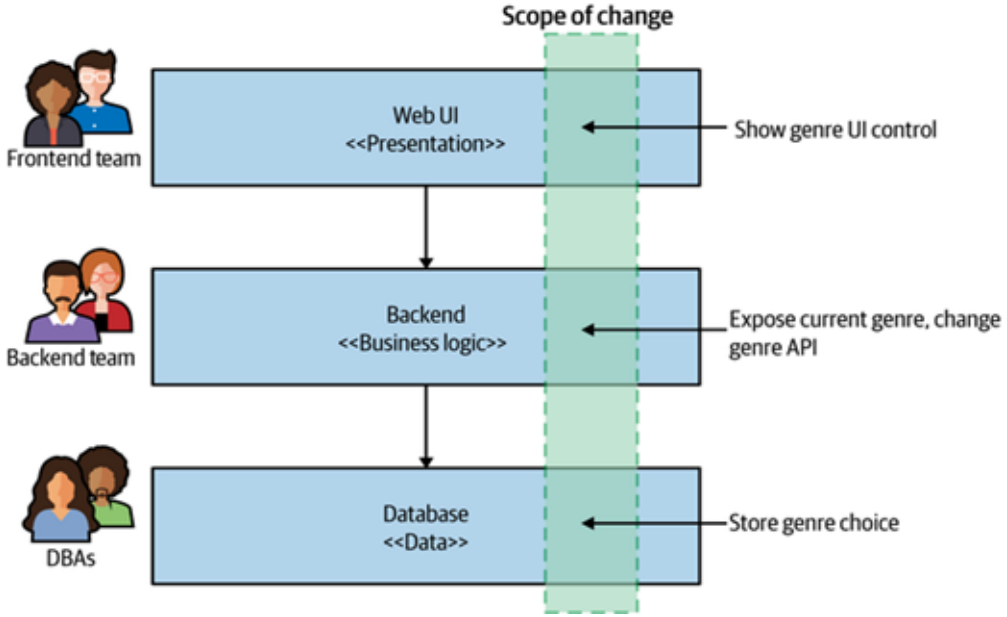
# Domain-driven modeling

- Model services around business capabilities

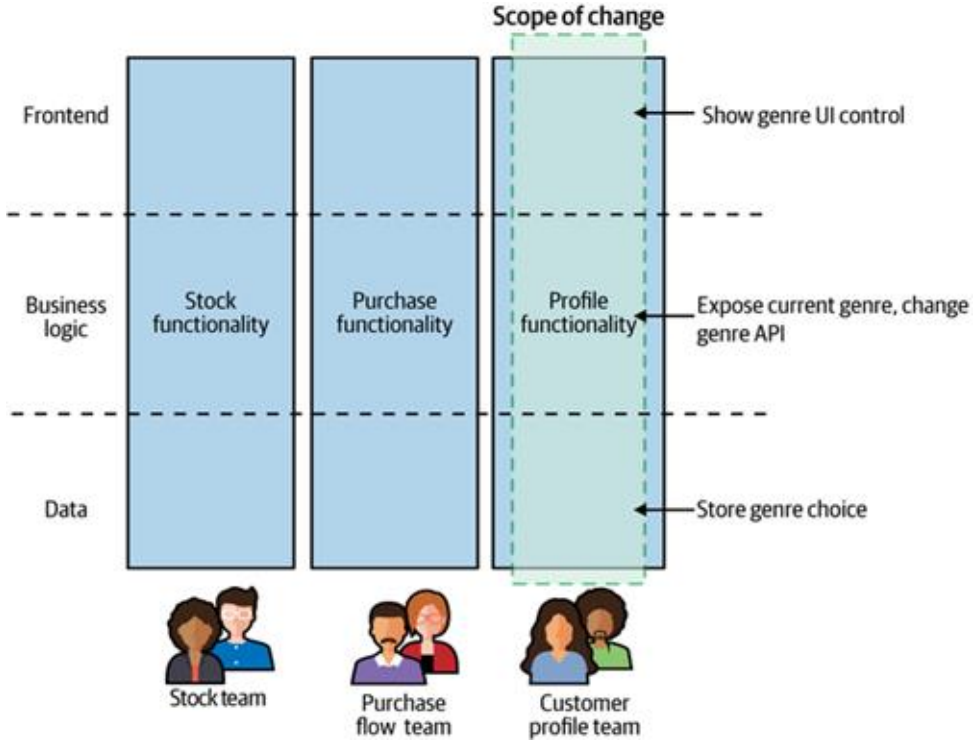


# Domain-driven modeling

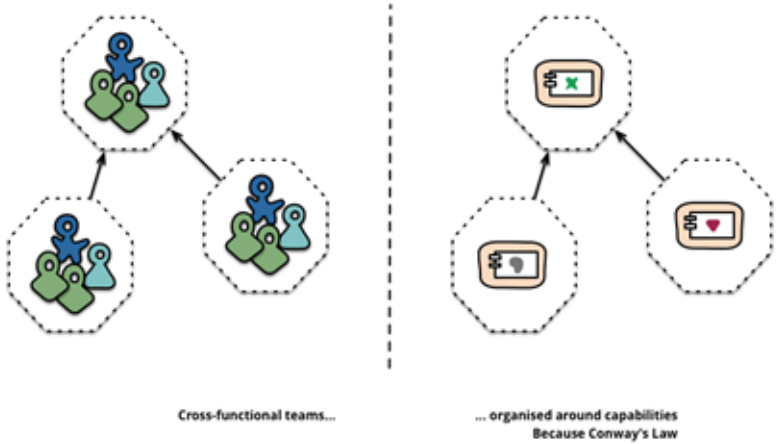


# Domain-driven modeling




## Remember Conway's Law?



Domain Driven Modeling

Culture of Automation

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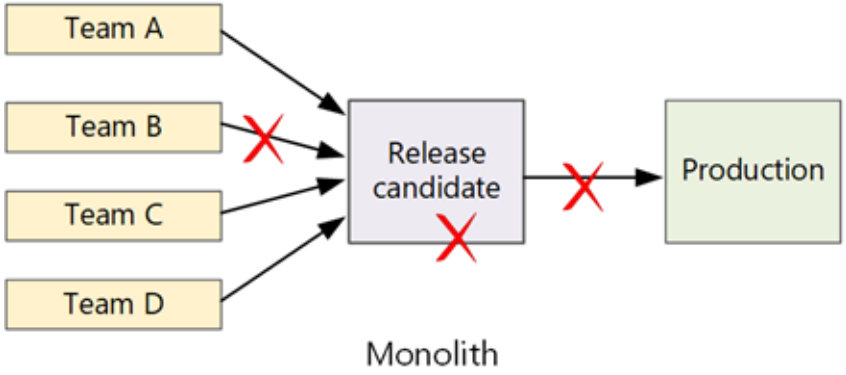
Sam Newman's Principles of Microservices



# Culture of Automation

- API-Driven Machine Provisioning
- Continuous Delivery
- Automated Testing

# Continuous Delivery



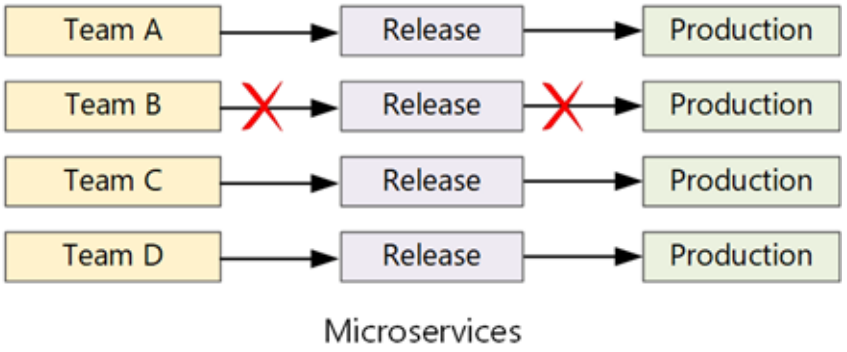




Image Source: https://learn.microsoft.com/en-us/azure/architecture/microservices/ci-cd

Domain Driven Modeling

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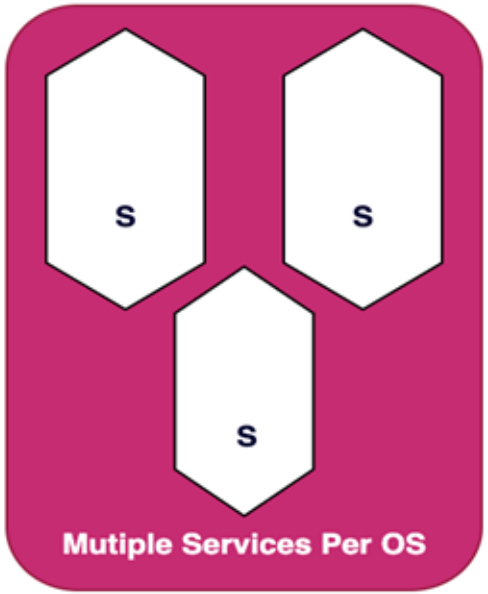

Sam Newman's Principles of Microservices

# Deploy Independently

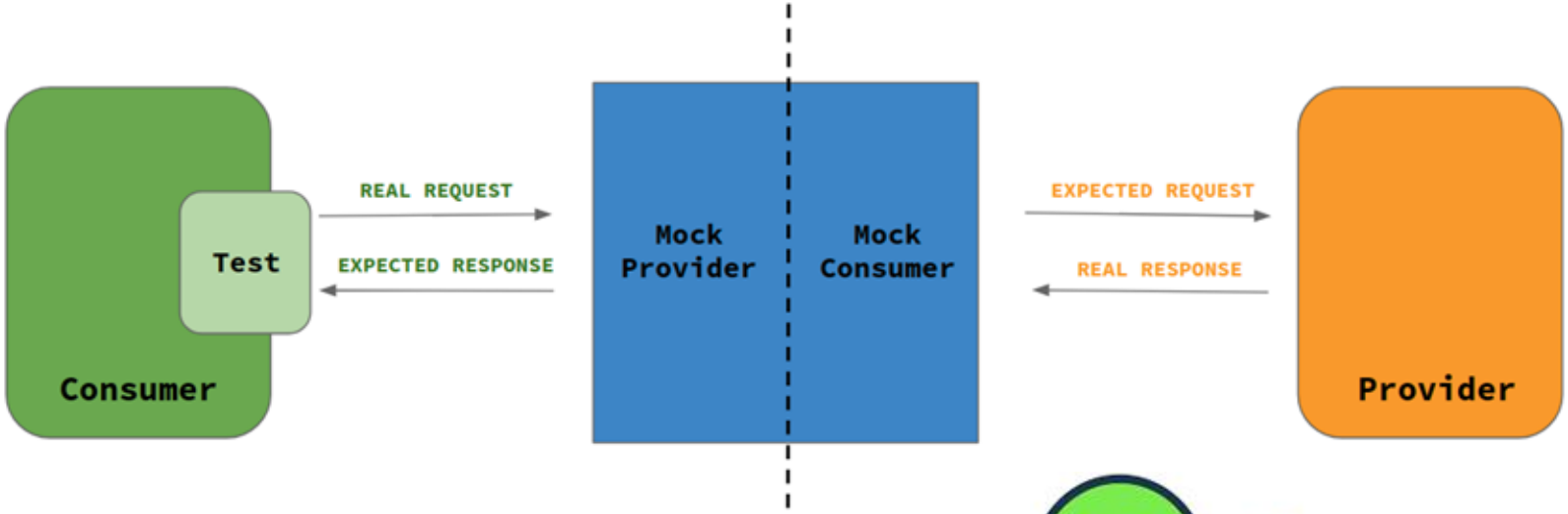
- One Service Per OS
- Consumer-Driven Contracts
- Multiple coexisting versions

# One Service Per OS



# Consumer-Driven Contracts

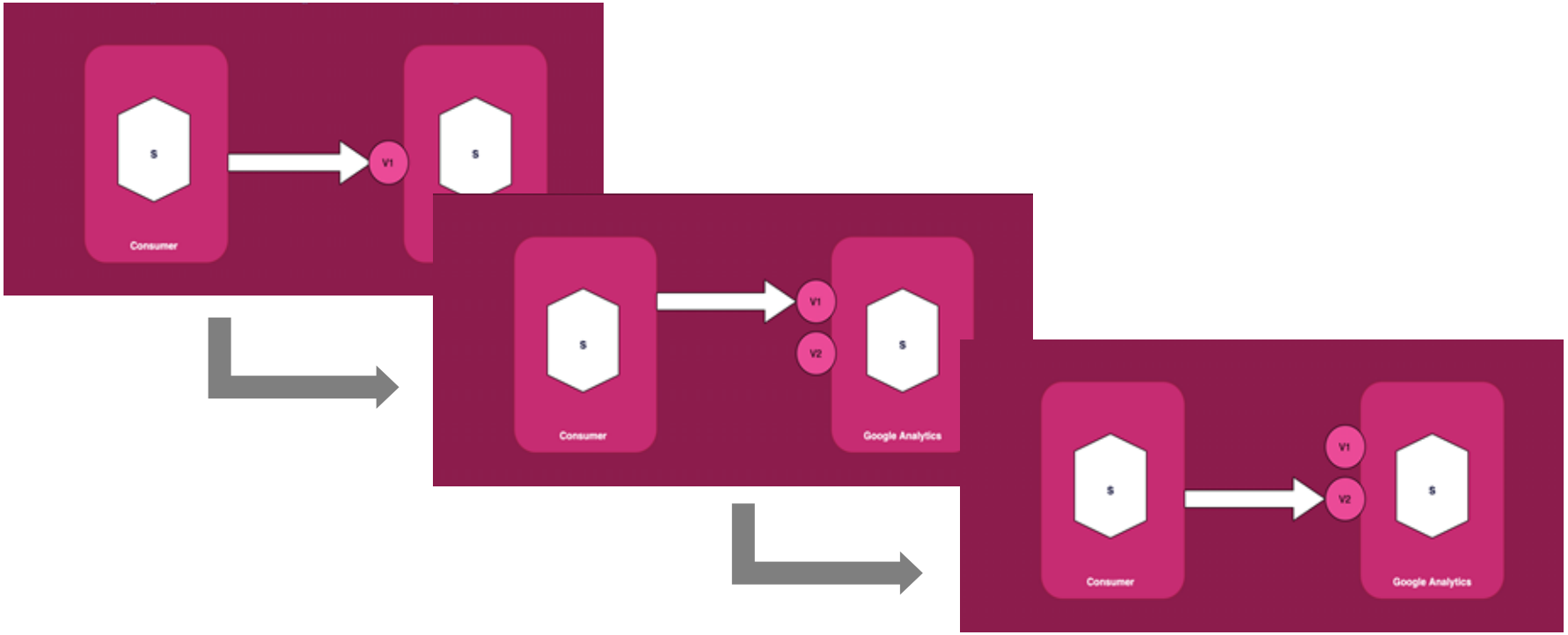






https://medium.com/@japneetkaur11/contract-testing-with-pact-17909b838de9

# Multiple coexisting versions



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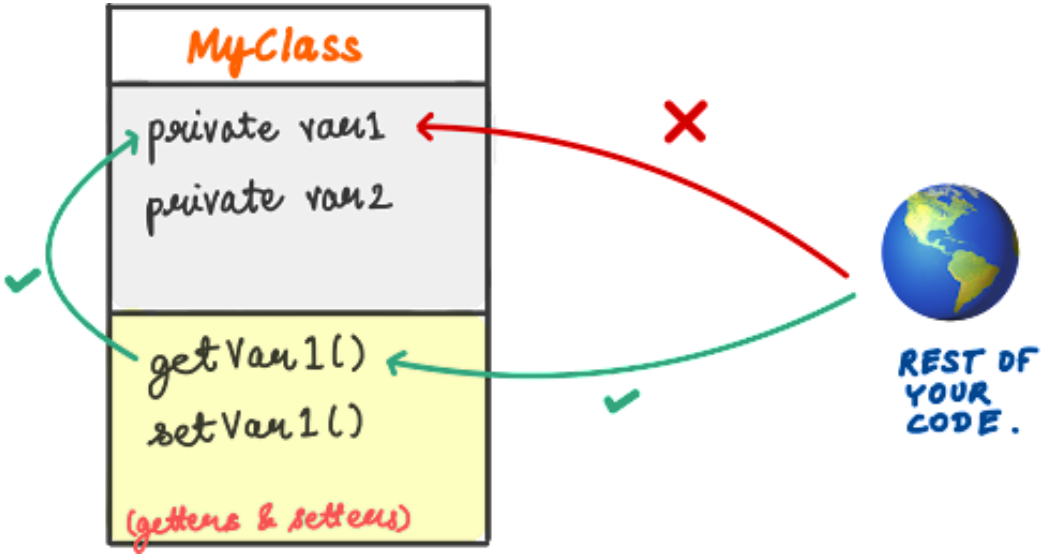


# Hide implementation details

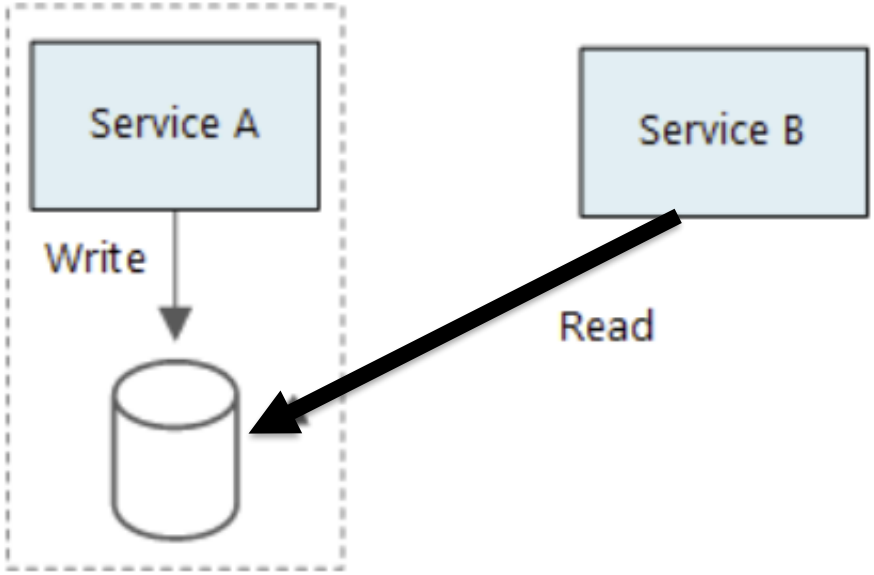
- Design your APIs carefully
- It’s easier to expose details later than hide them
- Do not share your database!

# Hide implementation details

Recall: Encapsulation in OOP



# Sharing database: Anti-pattern



Domain Driven Modeling

Culture of Automation

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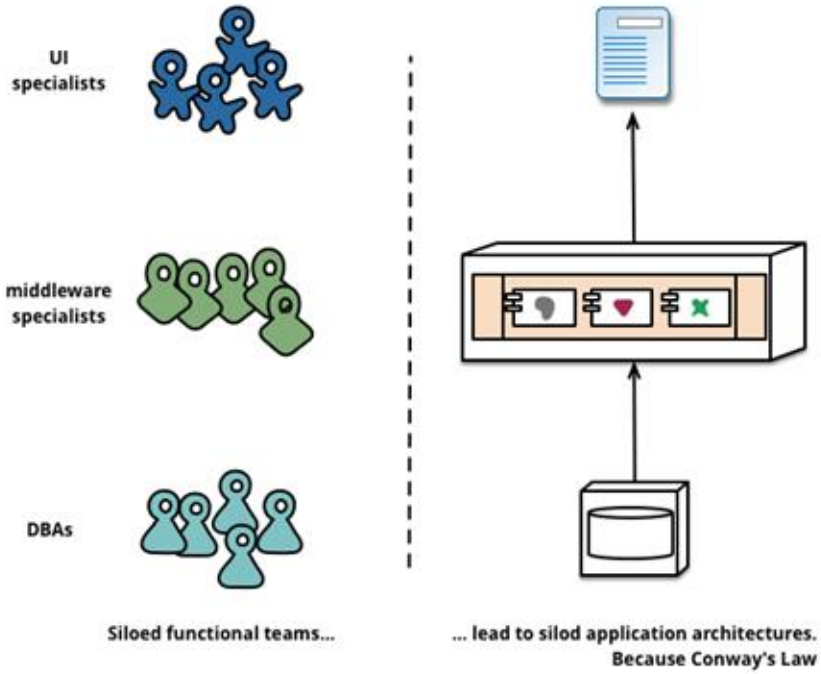
Sam Newman's Principles of Microservices

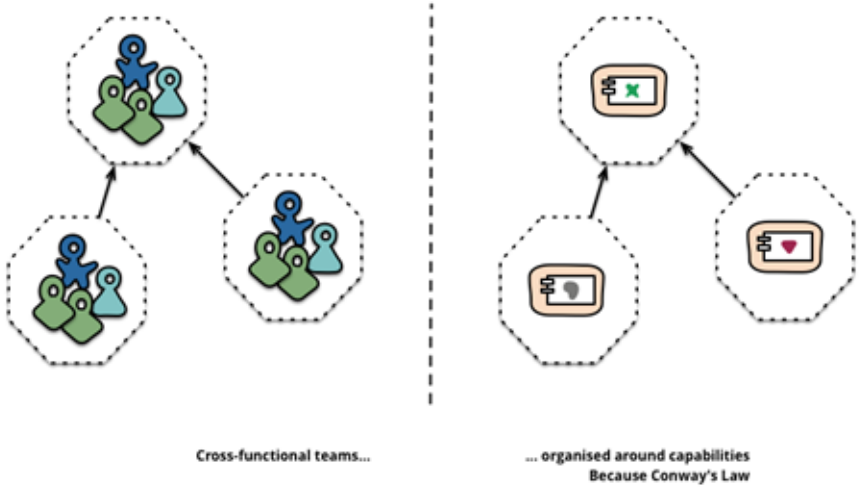


# Decentralized Governance

- Mind Conway's Law
- You Build It, You Run It
- Embrace team autonomy
- Internal Open Source Model

# Mind Conway's Law




"Products" not "Projects"

# YOU BUILD IT YOU RUN AWAY IT

"The traditional model is that you take your software to the wall that separates development and operations, and throw it over and then forget about it. Not at Amazon. You build it, you run it. This brings developers into contact with the day-to-day operation of their software. It also brings them into day-to-day contact with the customer. This customer feedback loop is essential for improving the quality of the service."

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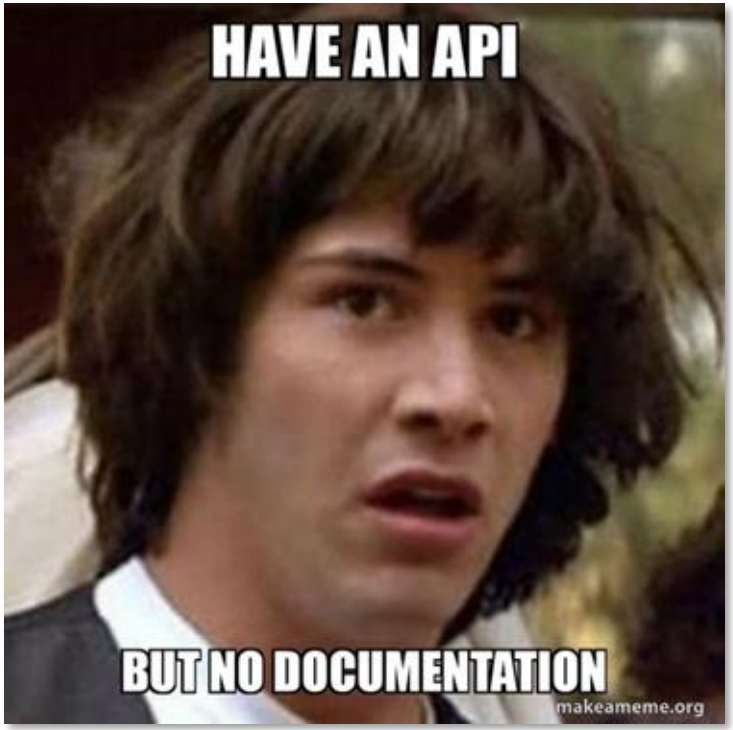
# Consumer First

- Encourage conversations
- API Documentation
- Service Discovery

# Encourage conversations



# Document your APIs!




Swagger™

Domain Driven Modeling

Culture of Automation

Hide Implementation Details

Decentralized Governance

Deploy Independently

Consumer First

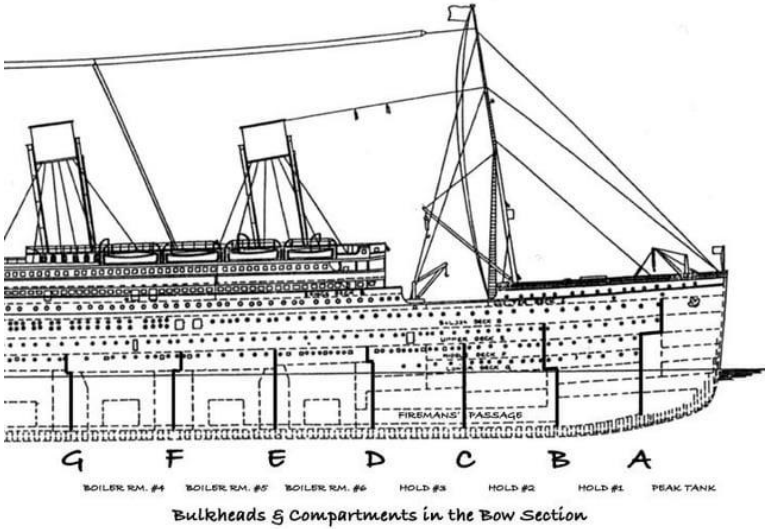
Isolate Failures

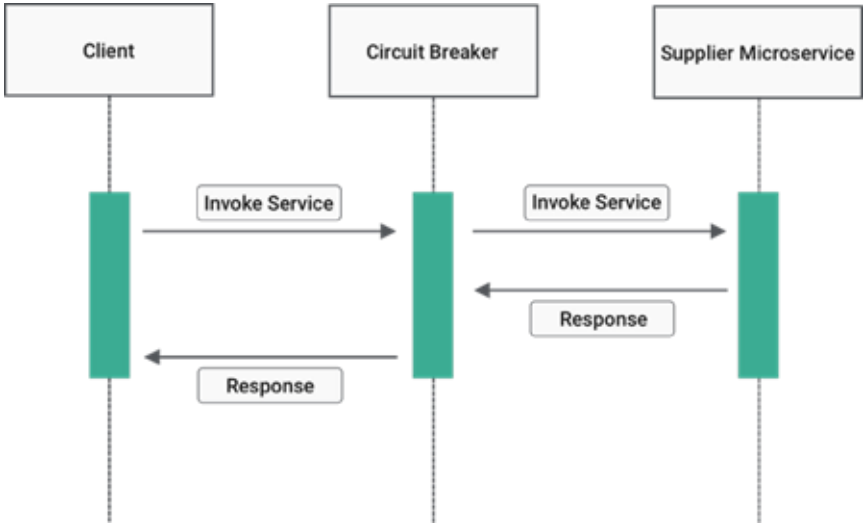



Sam Newman's Principles of Microservices

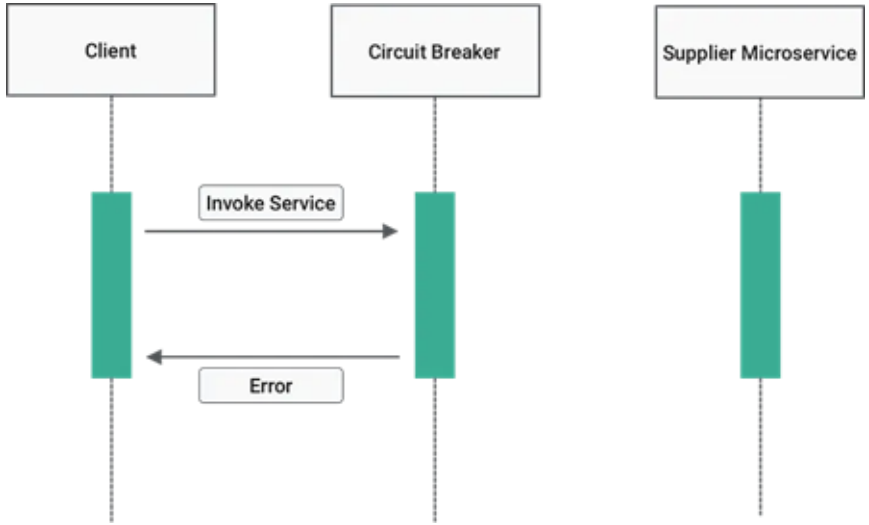
# Isolate Failure

- Avoid **cascading failures**
- Timeouts between components
- **Fail fast** aka *Design for Failure*
  - Bulkheading / Circuit breakers






Closed Circuit




Open Circuit

blogs.halodoc.io

# Are microservices always the right choice?

# Advantages of Microservices

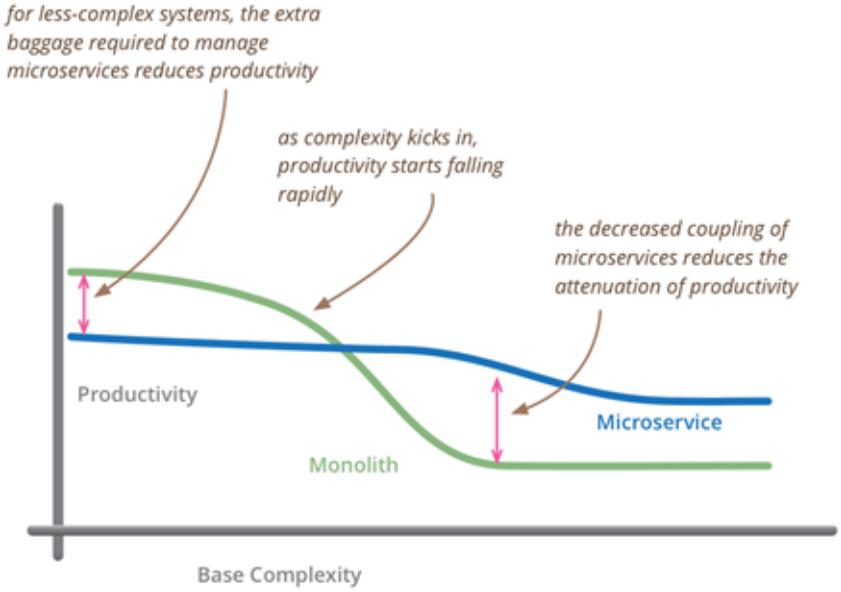
- Ship features faster and safer
- Amenable to horizontal scaling
- Target security concerns via isolation
- Allow the interplay of different systems and languages; no commitment to a single technology stack
- Easily deployable and replicable
- Embrace uncertainty, automation, and faults
- Better alignment with organization structure

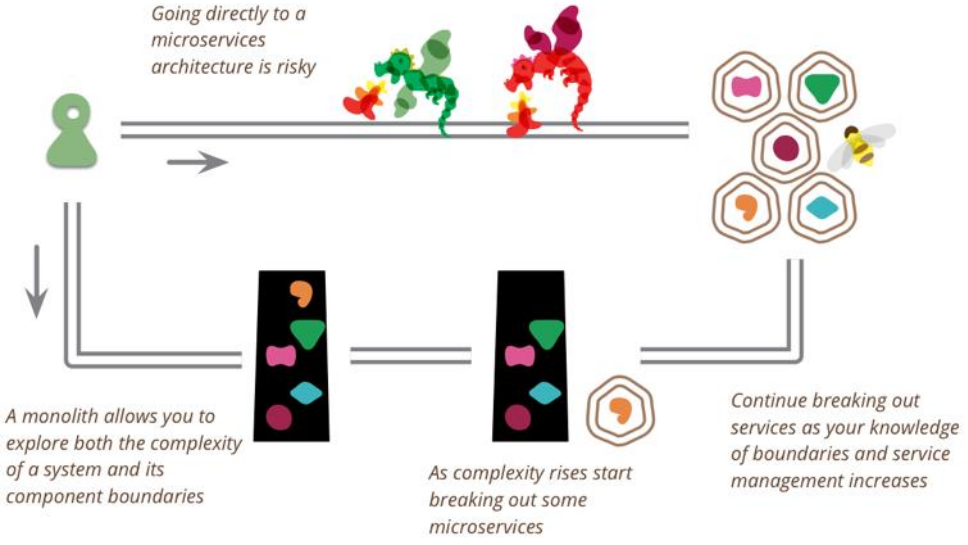
# Microservice Challenges

- Too many choices
- Lag between investment and payback
- Complexities of distributed systems
  - network latency, faults, inconsistencies
  - testing challenges
- Monitoring is more complex
- More system states
- More points of failure
- Operational complexity
- Frequently adopted by breaking down a monolithic application

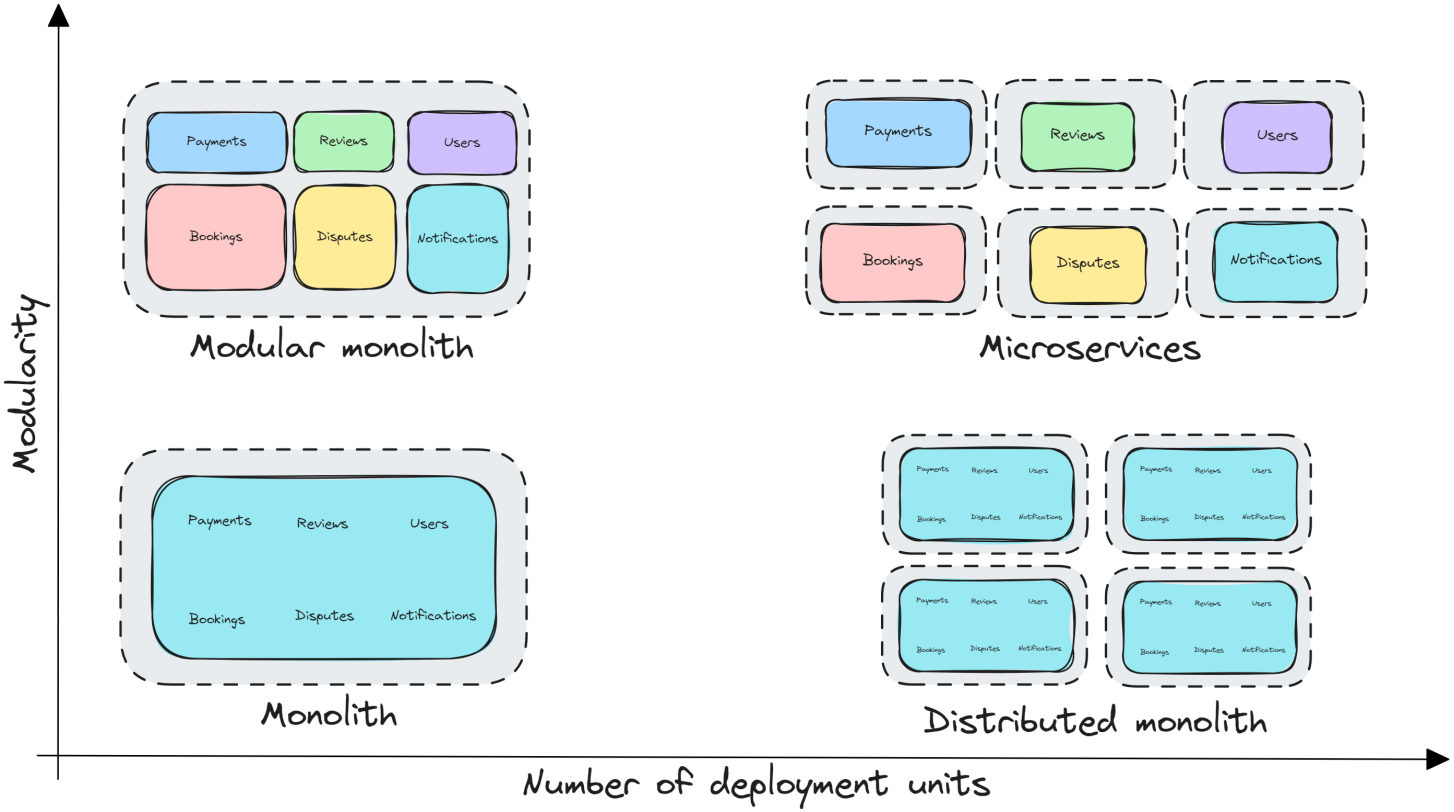


# Microservices Overhead




https://martinfowler.com/bliki/MonolithFirst.html




https://www.milanjovanovic.tech/blog/what-is-a-modular-monolith